DataDNA

Control - Trust - Act

Overview

Teradata.com/trusted-ai

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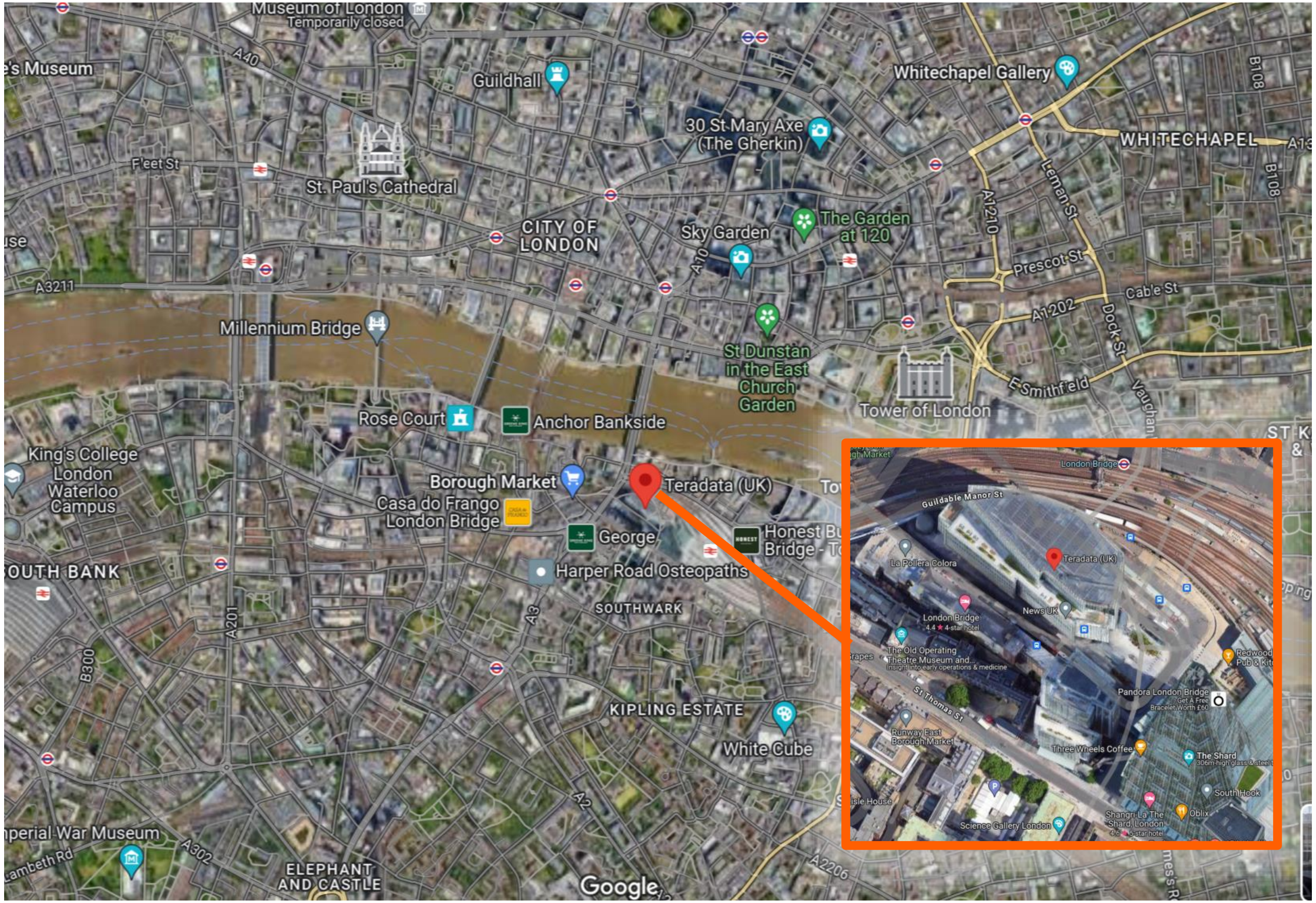
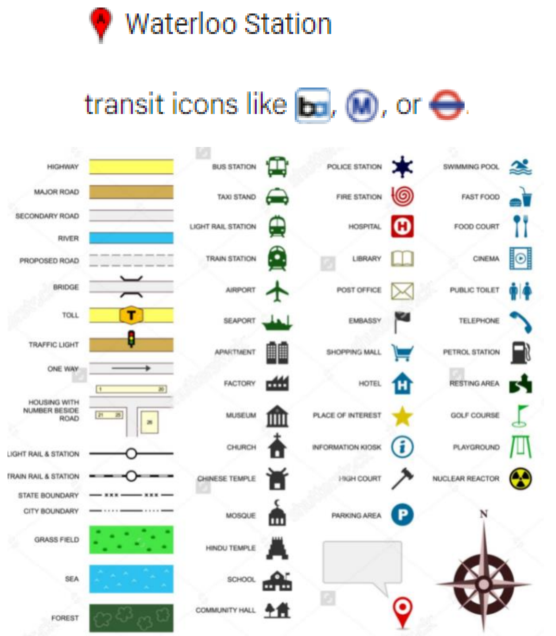
DataDNA The google map of your Eco-system





Point to Layers and click **More** to find info about:

- **Terrain:** Local terrain
- **Traffic:** Local traffic conditions
- **Transit:** Bus, subway, and rail routes
- **Biking:** Bike paths
- **Street View:** Street View of the area
- **Globe view:** Zoom out and view the world in 3D
- **Air quality:** Air quality information

Use cases driven by **Data**DNA insight

Reduce Cost, Risk, and Time to Value

Enable Trusted AI

Provide traceability for Feature Store, AI Models, Training Data that underpins AI Use Cases

Consolidation Simplification

Cleanup, consolidate Duplication Removal Candidates for replacement, migration or archiving

Migration & Modernisation

Quickly understand the effort, cost and risk of change, data mapping and trusted source

Compliance and Data Control

Data classification and sensitivity analysis for data governance and data control

Detailed Lineage for COREP Regulatory Reporting

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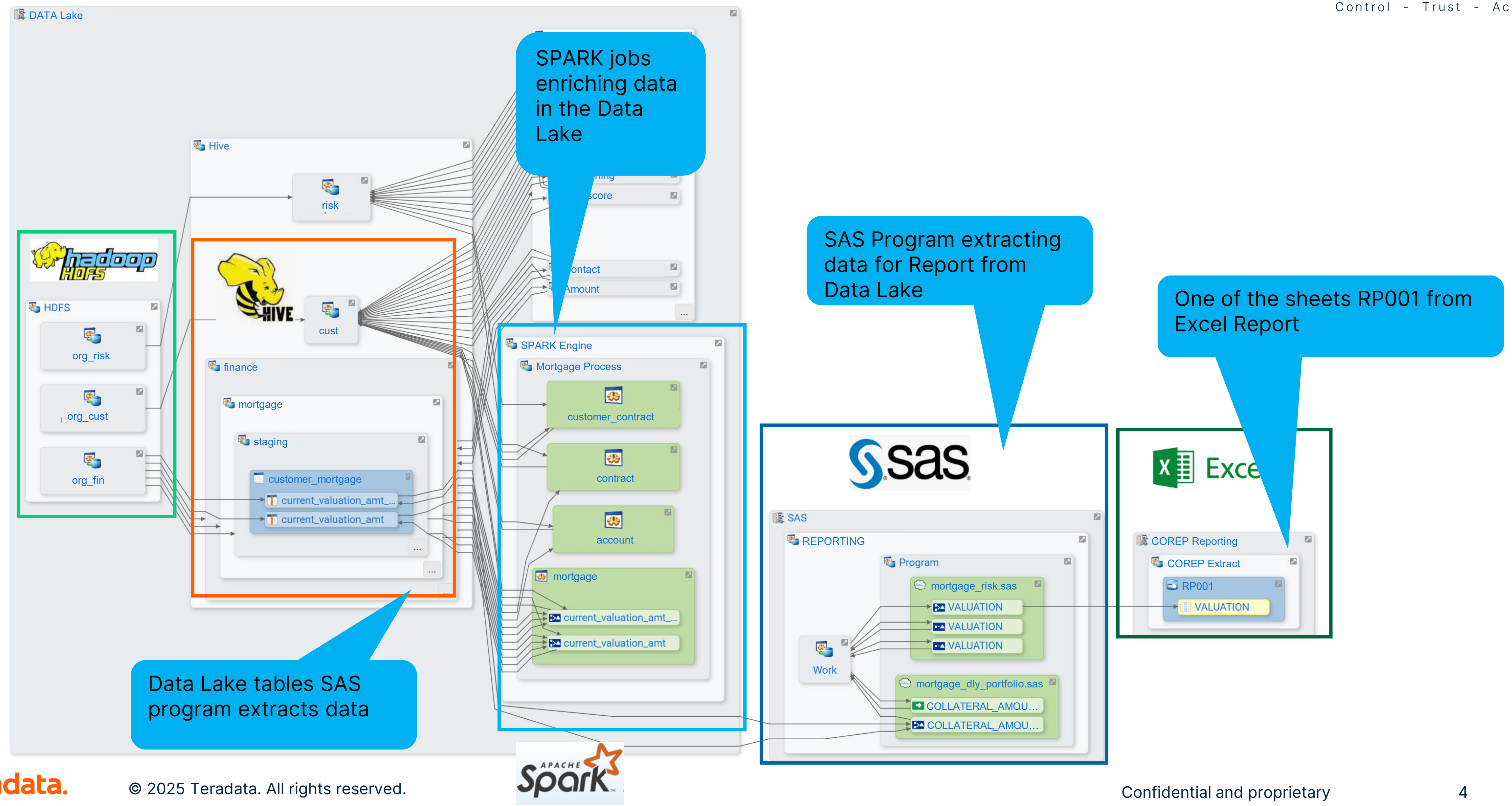

Fact Based Answers

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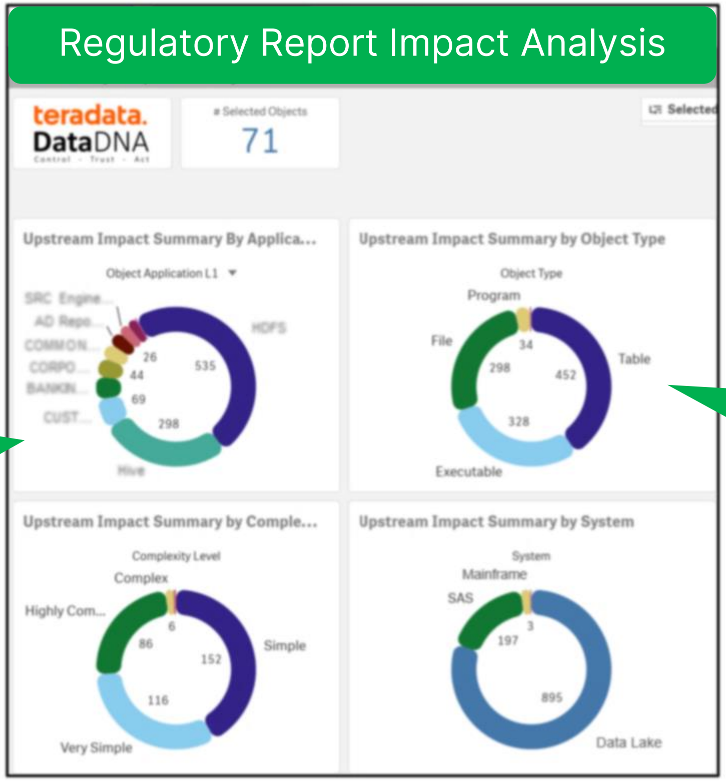
Results are not based on assumptions or craftsmanship, they are based on facts

Risks/Improvements Detection

Impact Analysis can now **easily ensure the data quality & control & efficiency** for the MIS, Statutory & Regulatory reporting, based in facts

Where does the data come from?

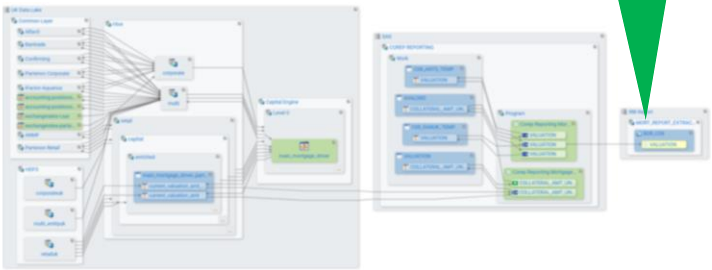


What type of objects are impacted

How complex are the calculations

What Systems are impacted

Lineage for Reporting Excel



Impact Analysis Example (Object Level)

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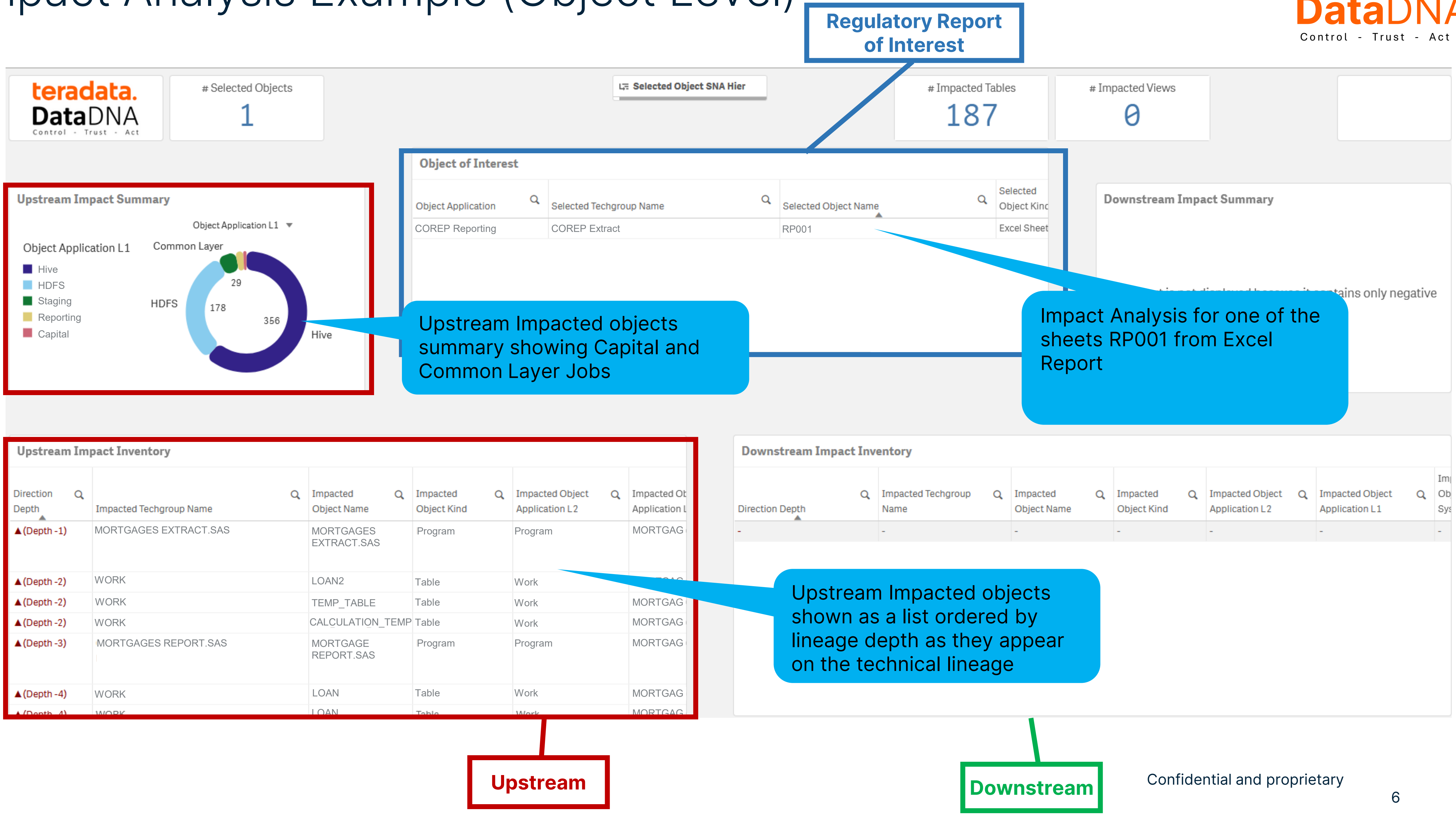

Impact Analysis Example (Column Level)

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Upstream Transformations Analysis

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#Transformations	# Distinct Transformations	#Distinct Input Columns
5	5	5

Upstream Column Lineage

Database Name	Impacted Table Database Object	Impacted Column Name	Direction Depth	Object Application L2
WORK	MORTGAGE_TEMP	VALUE_ADD	▲(Depth -2)	Work
WORK	CALCULATION_TEMP	VALUE_ADD	▲(Depth -2)	Work
WORK	LOAN	TOTAL_VAL	▲(Depth -4)	Work
WORK	LOAN2	NET_AMT	▲(Depth -6)	Work
WORK	VALUATION	COLLATERAL_AMOUNT	▲(Depth -6)	Work

Transform Input Sources

Database	Input Table	Input Column	Used on Condi...
FINANCE_MORTGAGE...	CUSTOMER_MORTGAGE	CURRENT_VALUATI...	N
FINANCE_MORTGAGE...	CUSTOMER_MORTGAGE	CURRENT_VALUATI...	N
WORK	LOAN2	TOTAL_VAL	N
WORK	LOAN	NET_AMT	N
WORK	EXTRACT	TOTAL_VAL	N
WORK	VALUATION	COLLATERAL_AMO...	N

Upstream Transformations & Associated Processes

Transform Type	Transformation	
SingleInput	<B>FROM PRC_SQL:</B> <B>FROM CREATETABLE#TEMP_TABLE:</B> <B>FROM SELECT#0:</B> SUM (CASE WHEN MARGIN_LEVEL = '3' THEN TOTAL_VALUATION ELSE 0 END) <FONT COLOR="#009900">- WHERE CONDITIONS: WHERE MARGIN_LEVEL = '3' AND ACCOUNTING = NULL </FONT>	
PassThru	<B>FROM BASE_STEP:</B> SET EXTRACT	...RTGAGE REPORT.SAS
MultiInput	<B>FROM PRC2_SQL:</B> <B>FROM SELECT#1:</B> CASE WHEN CURRENT_VALUATION_AMT_1 > 0 AND CURRENT_AVM_CONFIDENCE > 0 THEN CURRENT_VALUATION_AMT_1 ELSE CURRENT_VALUATION_AMT END AS COLLATERAL_AMOUNT <FONT COLOR="#009900">	MORTGAGE REPORT.SAS
MultiInput	<B>FROM PRC3_SQL:</B> <B>FROM SELECT#0:</B> MIN (COLLATERAL_AMOUNT, NET_AMT)	MORTGAGE REPORT.SAS
SingleInput	<B>FROM PRC4_SQL:</B> <B>FROM CREATETABLE#CALCULATION_TEMP:</B> <B>FROM SELECT#0... B> SUM (CASE WHEN MARGIN_LEVEL = '1' THEN TOTAL_VALUATION ELSE 0 END) <FONT COLOR="#009900">-WHERE ...NDITIONS: WHERE MARGIN_LEVEL = '1' AND ACCOUNTING != NULL </FONT>	MORTGAGES EXTRACT.SAS

COLLATERAL_AMOUNT is calculated from CURRENT_VALUATION_AMT in the Data Lake

VALUE_ADD is calculated as MIN of “Collateral Amount” and “Net amount”

With DataDNA – Business & Technical are unified

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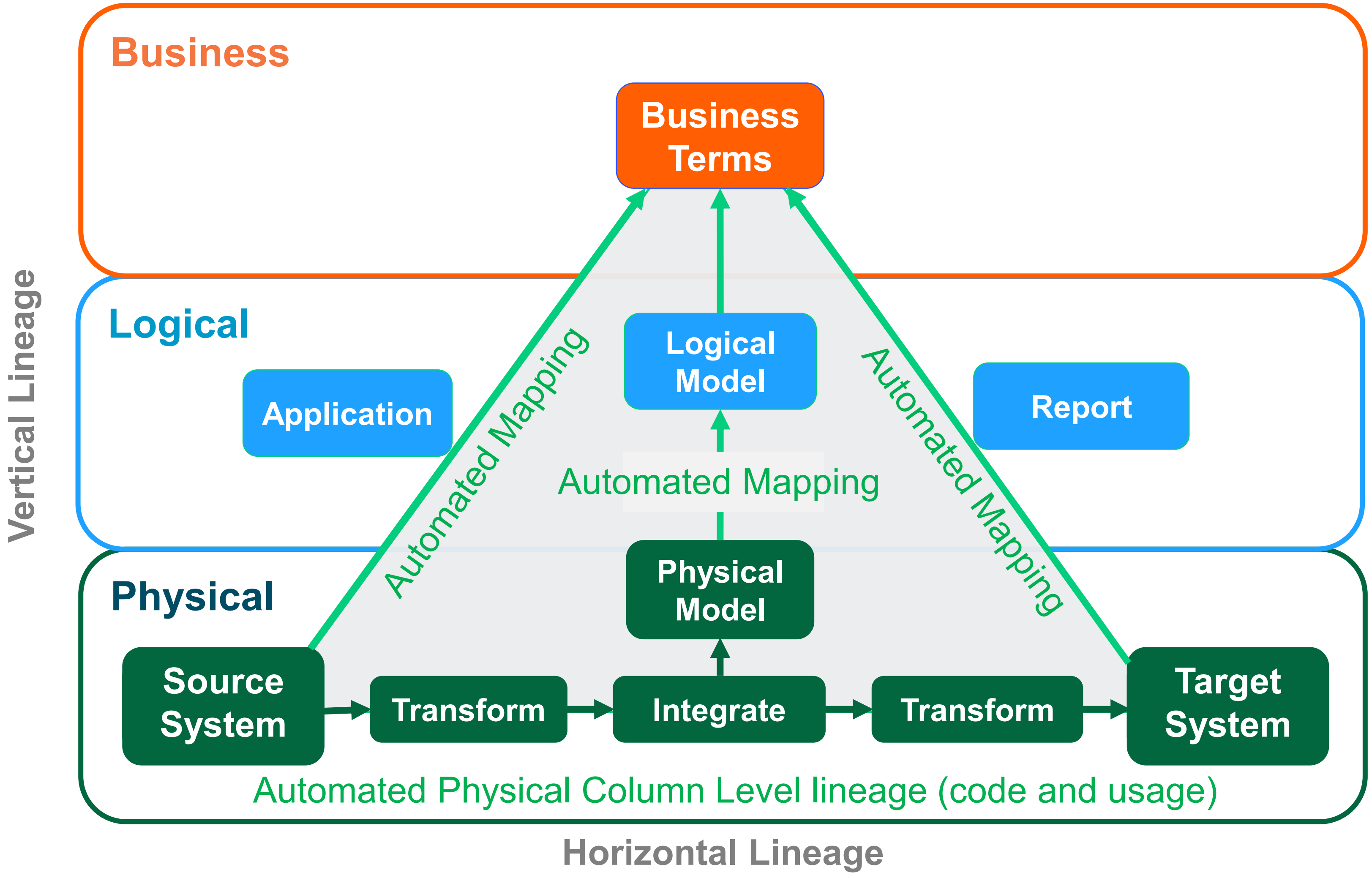

- **Vertical Lineage** - Business Terms mapped to physical data via Logical and Physical Models
- **Fact based logical Lineage** from physical lineage.
- **Fact based physical lineage** derived from **code** and system metadata
- **Fact based usage lineage** derived from **usage** logs
- **Vertical lineage mapping source and target data** to business terms is automatically derived saving a massive manual effort

With DataDNA – Business & Technical are unified

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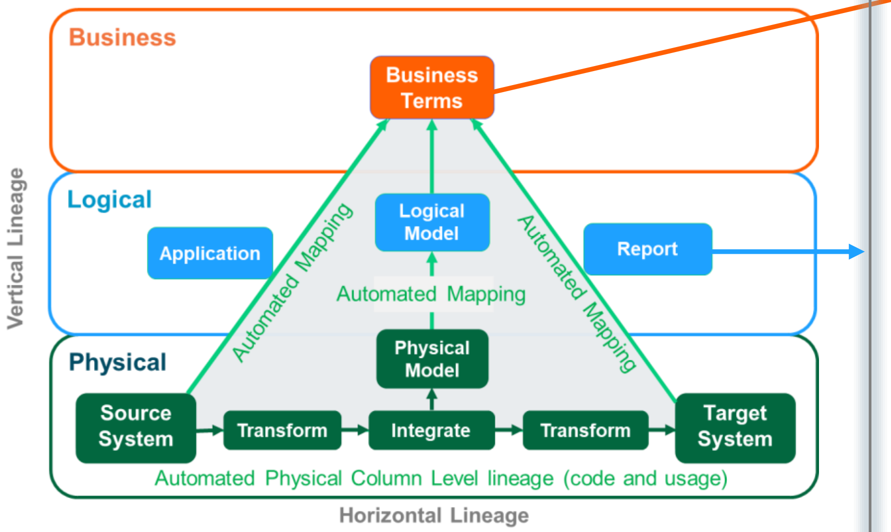
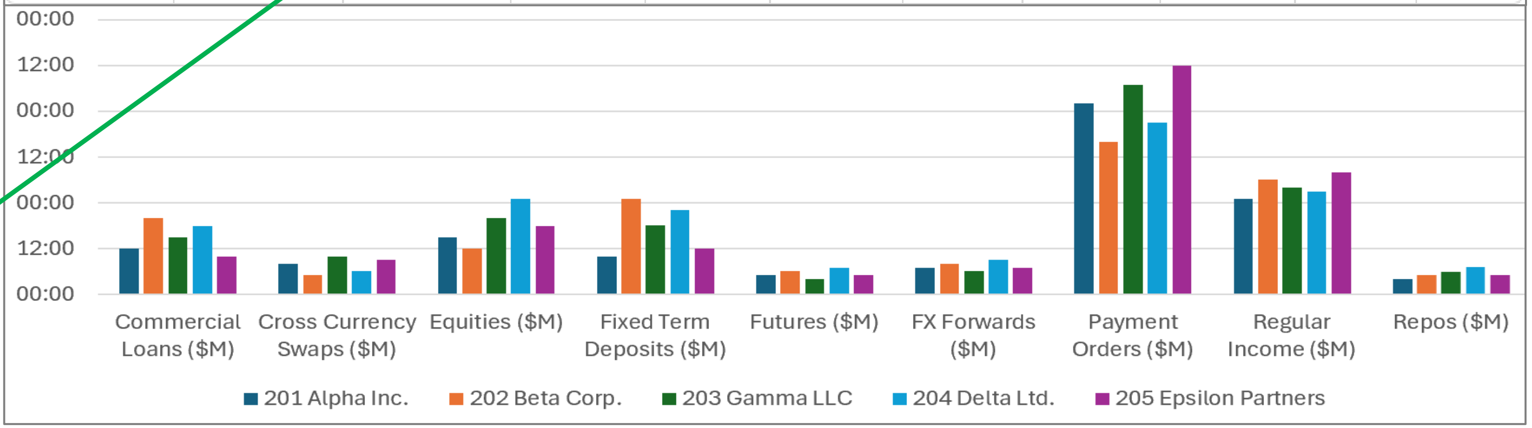
What's business definition?



Where does data come from?
(Horizontal Lineage)

Total Risk Report - TR1234

Corporation ID	Corporation Name	Commercial Loans ($M)	Cross Currency Swaps ($M)	Equities ($M)	Fixed Term Deposits ($M)	Futures ($M)	FX Forwards ($M)	Payment Orders ($M)	Regular Income ($M)	Repos ($M)
201	Alpha Inc.	12.5	8.3	15.2	10.0	5.0	7.2	50.0	25.0	4.5
202	Beta Corp.	20.0	5.5	12.0	25.0	6.5	8.0	40.0	30.0	5.0
203	Gamma LLC	15.3	10.0	20.0	18.5	4.0	6.5	55.0	28.0	6.0
204	Delta Ltd.	18.0	6.2	25.0	22.0	7.0	9.5	45.0	27.0	7.5
205	Epsilon Partners	10.0	9.0	18.0	12.5	5.5	7.0	60.0	32.0	5.5



One of thousands of Reports

Two Sides to the story, you must have BOTH

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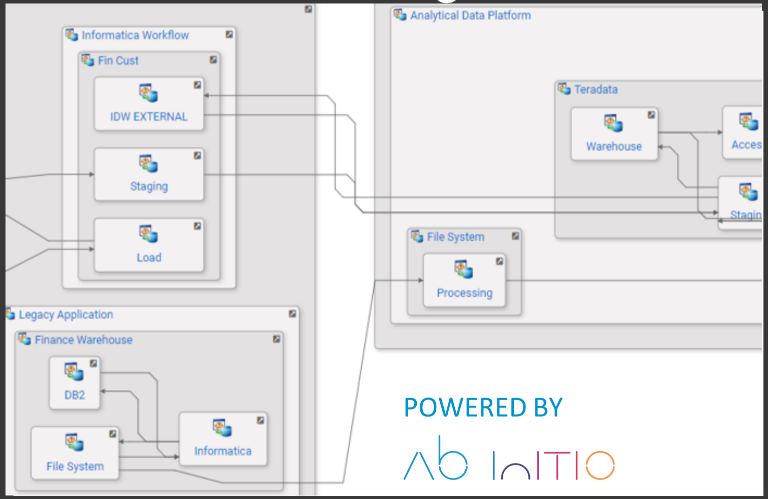
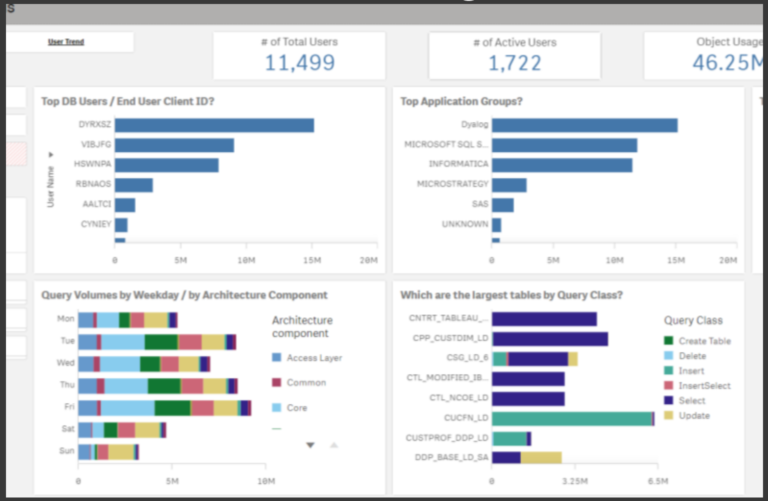
DataDNA Lineage Portal



Who uses which data?
Does it reconcile?

DataDNA Usage Portal



Where is data held?
How is it processed today?

DataDNA Live Demo

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Example Use Cases

DataDNA Use Cases

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Use Case	Sponsor	Description	Business Benefit	Customer Example
InDatabase Consolidation	CIO	Identify duplicate databases (typically where tactical sandpits have been created by the analyst community) in an analytics platform and develop remediation plan to consolidate into a strategic layer.	• Reduced operation cost through space saving (potential reduction in space of 10%+ • Reduced data wrangling by analyst community (currently spend 70% of time manipulating data)	**Enterprise Bank in EMEA** Decommissioning of c10 applications saved £13m/annum (conservative).
External Platform Consolidation	CIO	Identify duplicate BI platforms and consolidate onto a strategic analytics platform	• Reduced operation cost through platform decommissioning • Reduced data wrangling by analyst community (currently spend 70% of time manipulating data)	**Enterprise Bank in EMEA** An Enterprise MI platform (on DB2) has c80% data duplication with Teradata. Decommissioning the DB2 platform significant reduces IT cost.
Regulatory Compliance	CDO	Demonstrate end to end lineage and business glossary mapping based on how data is actually used and processed, required to meet regulatory requirements (e.g. BCBS239) to confirm business decisions are being made through trusted data.	• Reduce risk from proven fact-based compliance • Reduced cost through repeatable automated lineage and usage analytics	**Enterprise Bank in EMEA** Design & budget agreed for roll-out to c45 platforms, subject to commercials
Cost of Change	Various	Through accurate & automated data lineage, reduce the time & number of mistakes quantifying the impact of change on the analytical ecosystem (this is typically done manually which takes time and has low confidence in quality of output).	• Faster time to market	**Enterprise Manufacturer in EMEA** DataDNA reduced time for 1 impact assessment from ~600 hours to 1 minute, saving £8-10m per annum. Plus increased revenue of c£15m+/annum through new applications getting to market quicker.
Data Driven Decision Making	CEO	Through understanding usage of the analytical eco-system, accelerating the adoption of strategic projects (e.g. move to self-serve BI, reduced reliance on legacy software etc)	• Faster time to market • Faster time to analytics/decisions • Faster BizDevOps • Automation of critical processes	WIP
Expose Current Usage	Various	Identification of operational processes supported by unknown or undocumented data processing	• Reduced operational risk	**Enterprise Bank in EMEA** Identified processes across multi-platforms, previously undocumented

DataDNA Use Case Examples

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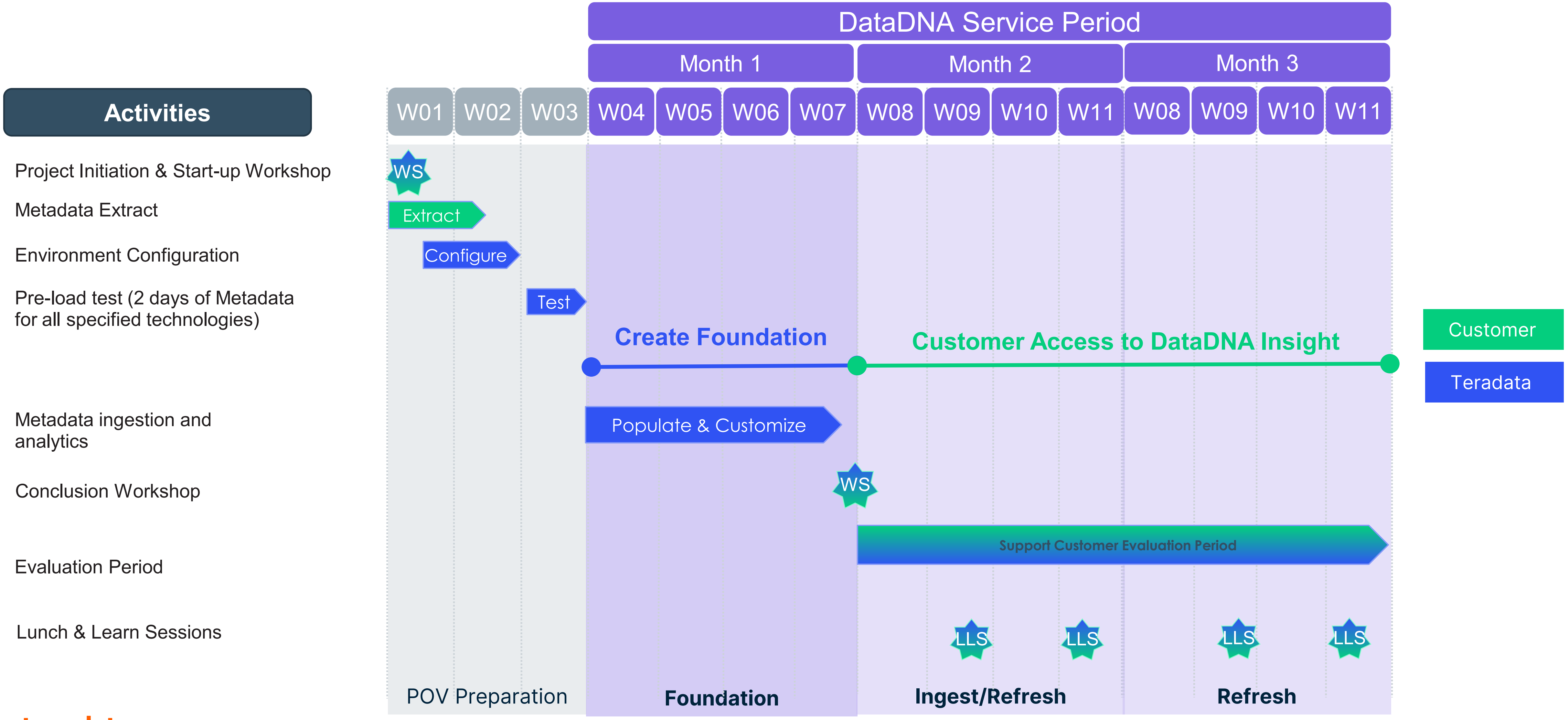
Examples taken from the following enterprise DataDNA enterprise customers;
Banks (4), Manufacturers (1), Retails (1), Telco (3)

Total Cost of ownership	• Free up Space - Teradata, Hadoop, Oracle, DB2 ... any platform • Business Optimisation of Data across all platforms
Data Governance	• Regulatory [BCBS 239, EUDRA, GDPR, etc.] • Reconciliation across platform(s), informed and driven by lineage
Integrating BI Estate	• Build analytics solution [any platform] as an integrated data management architecture • Analysis/planning for federated queries e.g. QueryGrid [Hadoop/Teradata] • Queryable Archive – highly compressed, cost effective bulk storage [petabytes], typically Hadoop but any platform • Reverse Engineering to re-platform legacy business apps e.g. mainframe apps automatically re-written to run on Hadoop
Application Integration	• Rapid Data Provisioning, Build & Operate [any applications on any platforms] – “reduce time to value” • Data Migration & Decommissioning, Simplification e.g. Legacy DW/Marts • Test Data Creation – any platform, any application

DataDNA Engagement Timeline (3 Months)



Preliminary Consolidation Planning

High Level Perspectives / Ideas

Phase 1

One-off clean-up Legacy , Group & EDW

- Quarantine identified unused data objects and associated ETL / BI
- Quarantine identified duplicate data objects and associated ETL / BI
  - Repointing (ETL/BI) required when used duplicate candidate is removed!

Phase 2, 3 etc

Consolidate remainder Objects/ ETL, options:

a) Legacy GDW*: per data facet(s)*
   - *e.g. Account, then CIS Customer etc, based on highest usage, cost, value, other?*

b) Group GDW*: per subject area(s)*
   - *e.g. BFS, then RBB P&O etc, based on highest usage, cost, value, other?*

c) Legacy / Group GDW: per *organisation(s)*
   - *e.g. Risk, then Group Finance etc, based on highest usage, cost, value, other?*

d) Legacy GDW*: combination of (a) & (c)*
   - *e.g. start with data facets for Risk, then Finance*

e) Group GDW*: combination of (b) & (c)*
   - *e.g. start with subject areas for Risk, then Finance*

Group GDW

1082 Active Business Users using 30 databases and 30384 active objects

- RISK (96 users, 16 databases, 22730 used objects – 7015 used tables)
- GROUP FINANCE (106 users , 20 databases, 656 used objects – 166 used tables)
- FINANCE (114 users, 20 databases, 967 used objects – 249 used tables)
- DATA PLATFORM (235 users, 26 databases, 2870 used objects – 788 used tables)

Common Objects GROUP FINANCE, FINANCE & RISK: 9 db and 273 objects

EDW

1252 Active Business Users using 49 databases and 8650 active objects

- BUS & EVRY DAY BNKG (38 users, 3 databases, 17 used objects – 7 used tables)
- STRATEGY (37 users, 10 databases, 1769 used objects – 1383 used tables)
- BUSINESS (64 users, 16 databases, 4188 used objects – 2007 used tables)
- FIN CRIME & FRAUD (34 users, 7 databases, 388 used objects -184 used tables)

Common Objects: … db, … objects

Legacy GDW

1006 Active Business Users using 39 databases and 11254 active objects

- RISK (55 users, 24 databases, 6535 used objects – 3179 used tables)
- CREDIT RISK (34 users, 23 databases, 7222 used objects – 3459 used tables)
- BUSINESS (63 users, 20 databases, 6348 used objects – 2994 used tables)
- CONSUMER DIVISION (43 users, 19 databases, 575 used objects – 259 used tables)

Common Objects: RISK, CREDIT RISK, BUSINESS & CONS. DIV: 13 db, 364 objects

Group Leads

440 Active Business Users using 24 databases and 2073 active objects

- DATA PLATFORM (206 users, 22 databases, 1685 used objects – 784 used tables)
- STRATEGY (31 users, 14 databases, 235 used objects – 136 used tables)
- BUSINESS (34 users, 14 databases, 196 used objects – 106 used tables)
- FINANCE (19 users, 12 databases, 251 used objects -.123 used tables)

Common Objects:7 db and 18 objects

Agile Data Labs

889 Active Business Users using 165 databases and 34274 active objects

- DATA PLATFORM (263 users, 71 databases, 11688 used objects – 11122 used tables)
- BUSINESS (60 users, 32 databases, 10062 used objects – 9850 used tables)
- CONSUMER BANK (20 users, 8 databases, 551 used objects – 542 used tables)
- FINANCE (72 users, 30 databases, 2489 used objects – 2327 used tables)

Common Objects: DATA PLATFORM, BUSINESS, CONS BANK, FINANCE: 1 db, 1 object

Phase 2 Option C Initial Analysis

INFORMATICA / TPT

Group GDW

Duplicate Table #	Primary Table #	Ratio
4530	713	0.12

9 TB

Active Objects	#	Incl dropped	Used Active Objects	#	Incl dropped	Unused Active Objects	#	Incl dropped
Total	181045	206802	Total	33771	59528	Total	147274	147274
Tables	18409	42084	Tables	9958	33633	Tables	8451	8451
Views	162636	164718	Views	23813	25895	Views	138823	138823

5 TB

373

32951

19033

250

INFORMATICA / TPT/ MULTILOAD

EDW

EDW Active Objects	#	Incl dropped	Used Active Objects	#	Incl dropped	Unused Active Objects	#	Incl dropped
Total	39675	50872	Total	24626	35819	Total	15049	15053
Tables	24141	35337	Tables	16491	27683	Tables	7650	7654
Views	15534	15535	Views	8135	8136	Views	7399	7399

DATASTAGE / MULTILOAD INFORMATICA

Legacy GDW

Duplicate Table #	Primary Table #	Ratio
37330	1420	26.29

2.5 TB

Active Objects	#	Incl dropped	Used Active Objects	#	Incl dropped	Unused Active Objects	#	Incl dropped
Total	145234	150200	Total	28172	33135	Total	117062	117065
Tables	121549	126510	Tables	21740	26698	Tables	99809	99812
Views	23685	23690	Views	6432	6437	Views	17253	17253

5 TB

823

717

Phase 1 Initial Analysis

Source Code Lineage

DataDNA Lineage portal

DataDNA Source Code Lineage

The real picture of how systems are working

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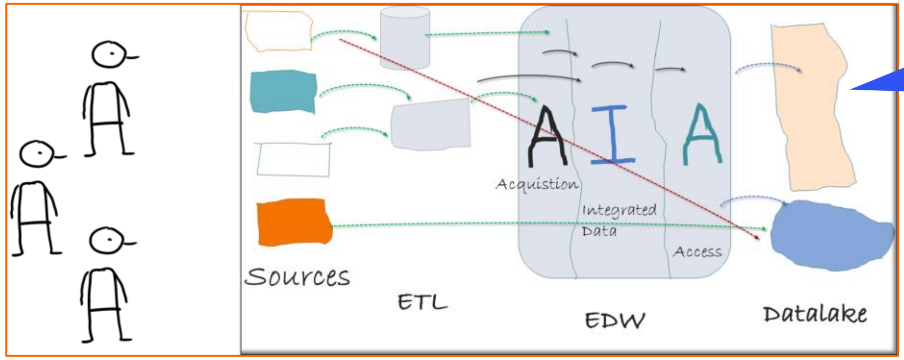
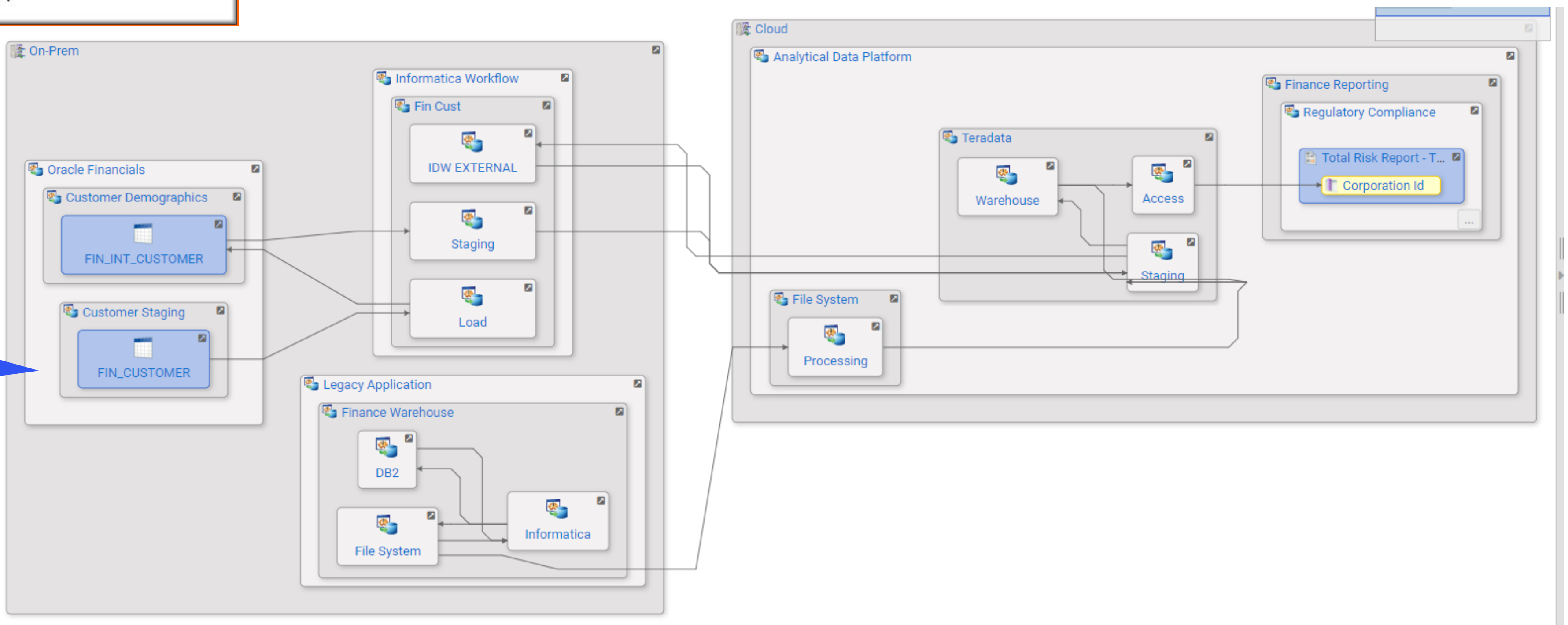


How the system was originally designed

DataDNA shows how systems are actually built



DataDNA – High-level Application Lineage

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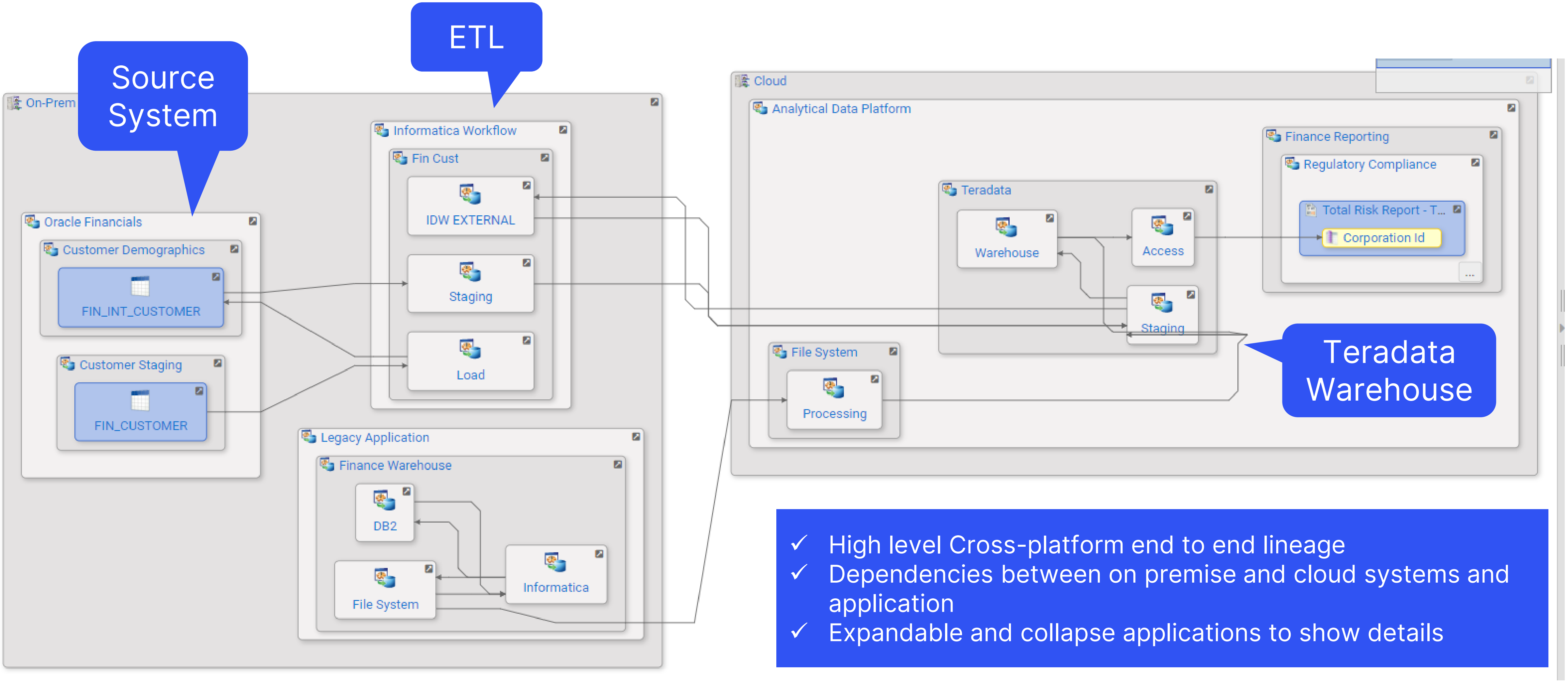

- ✓ High level Cross-platform end to end lineage
- ✓ Dependencies between on premise and cloud systems and application
- ✓ Expandable and collapse applications to show details

Table level Lineage

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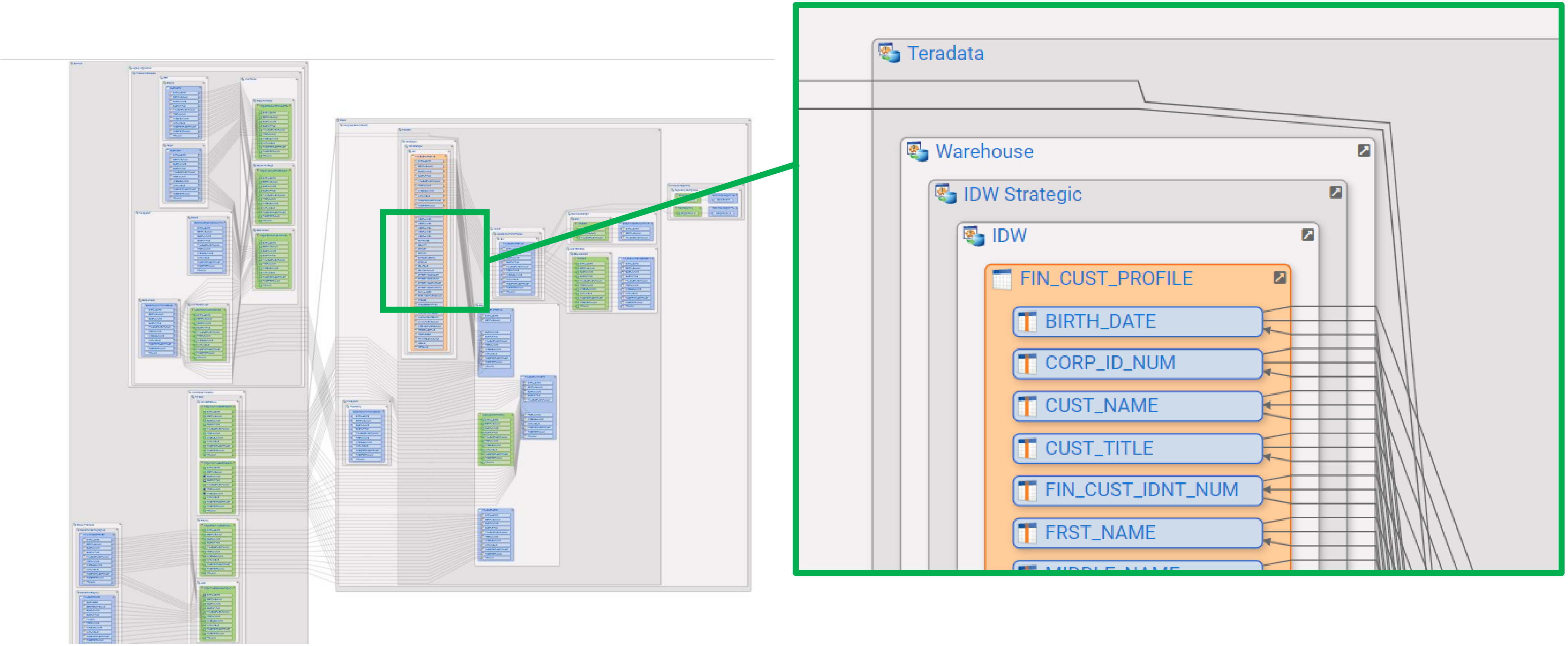

- ✓ Table and view level lineage (shown in blue)
- ✓ Inputs and output to programs that process and transform data (shown in green)
- ✓ One interface regardless of underlying technology

DataDNA Detailed Column Level Lineage

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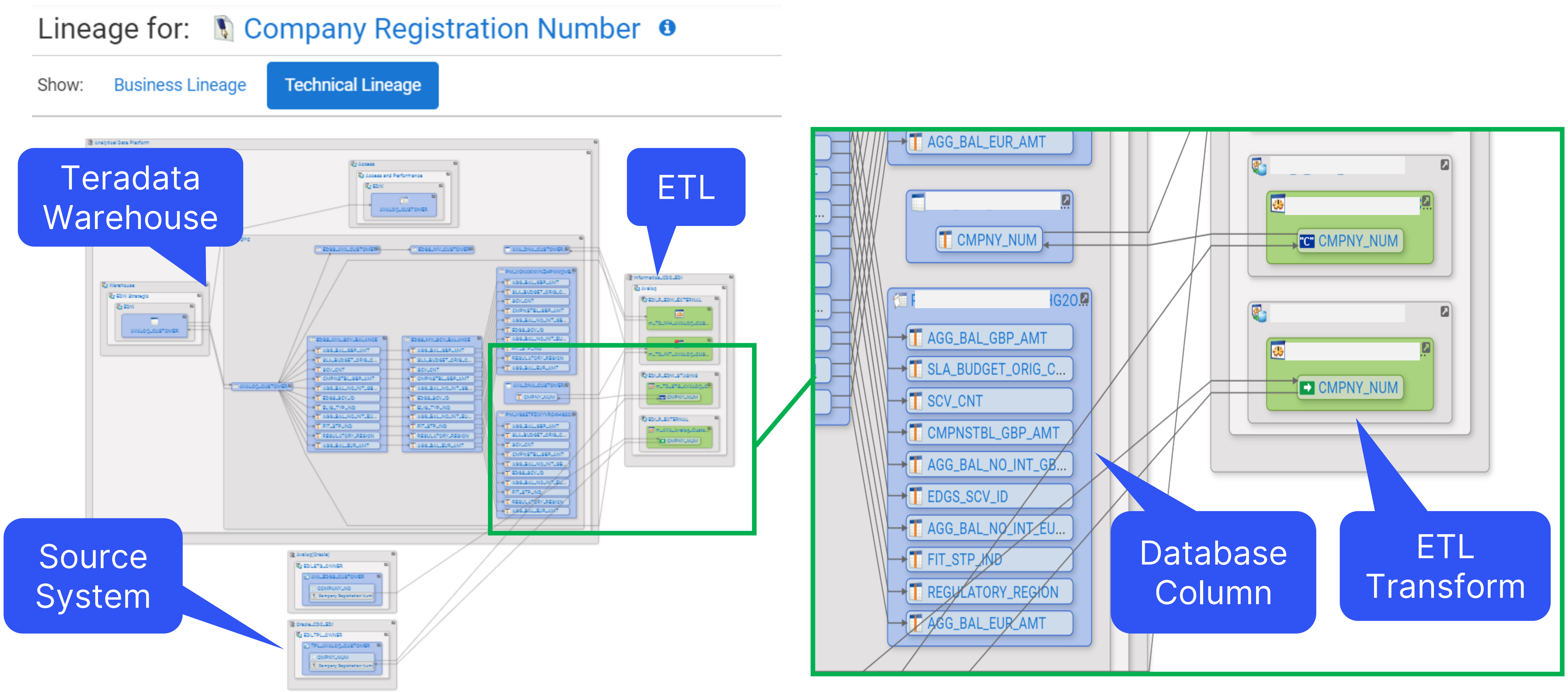

DataDNA - Detailed Transformation logic

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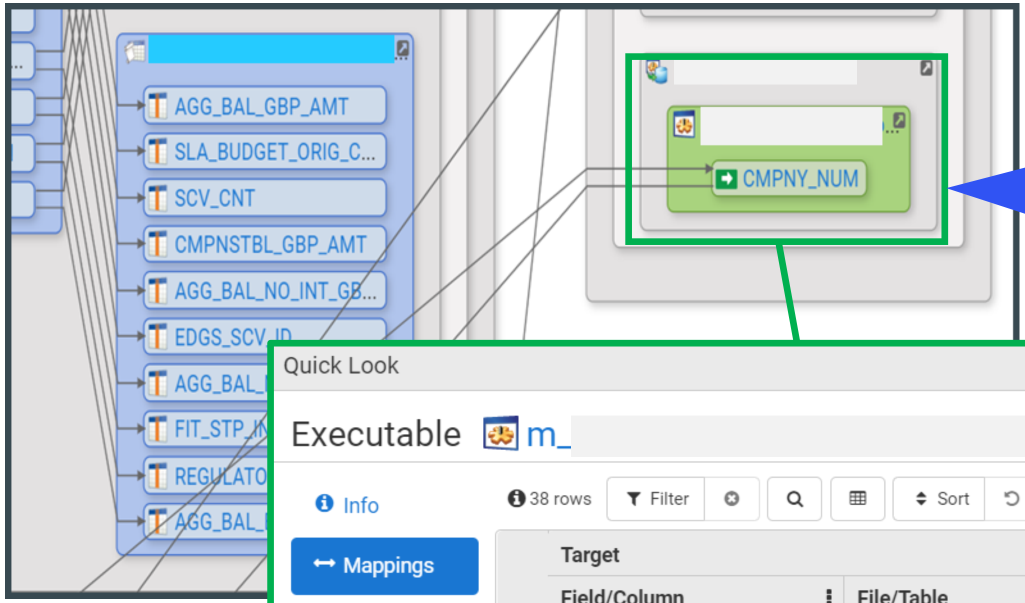
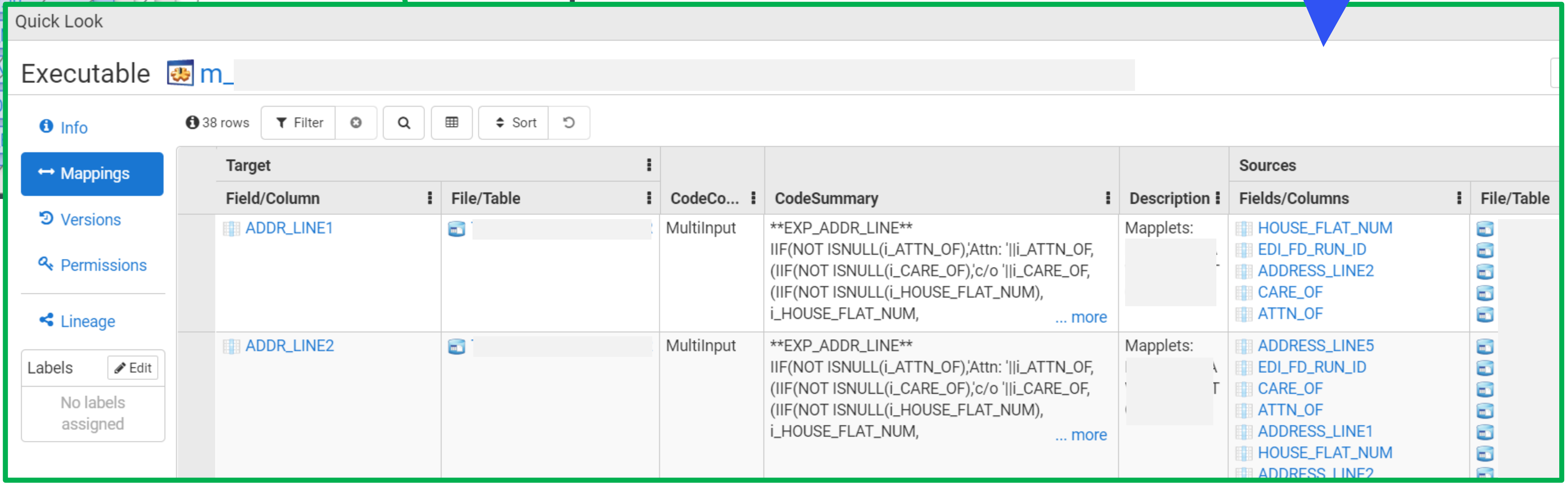
DataDNA automatically parses Code of different languages to derive end to end lineage

Detailed Transforms logic and mapping





Usage Lineage

DataDNA Lineage portal

DataDNA Uses Lineage
Derived from the actual queries users run!

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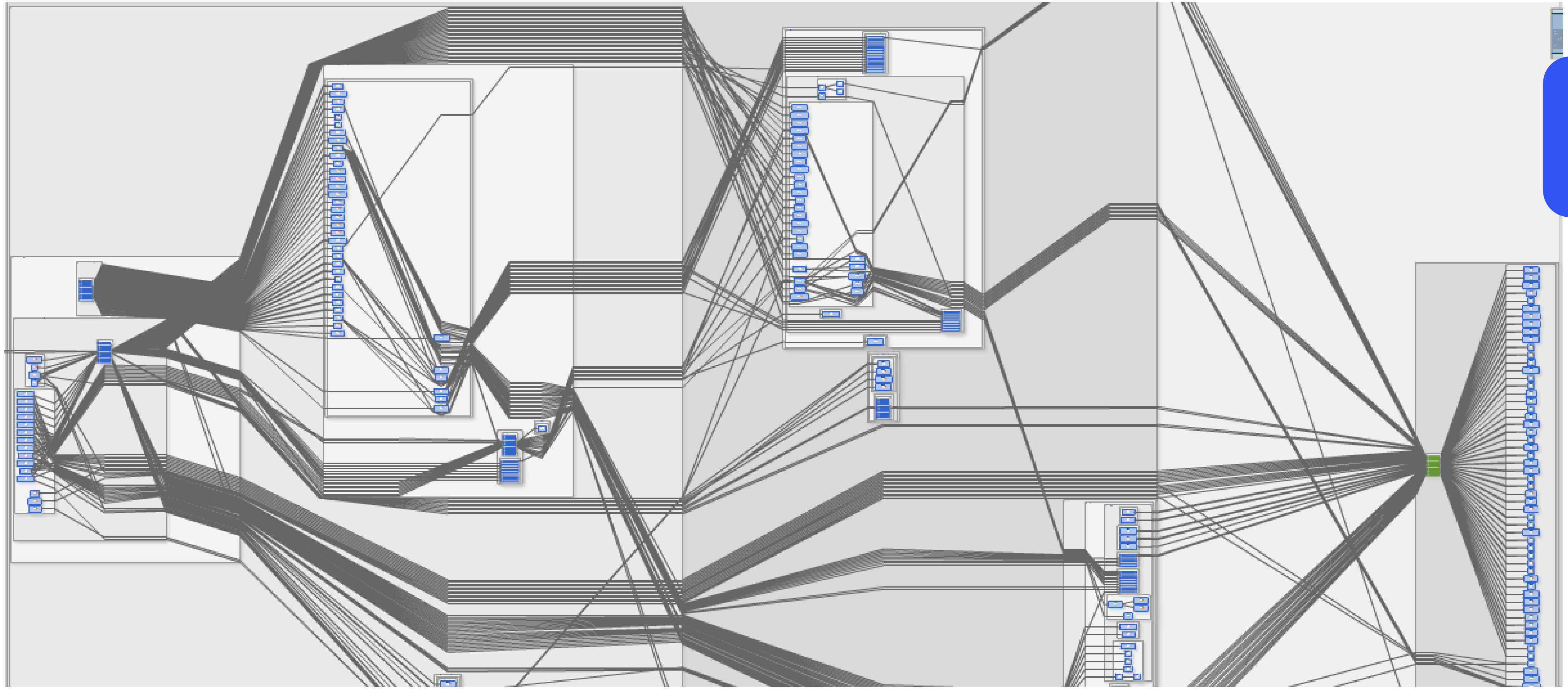

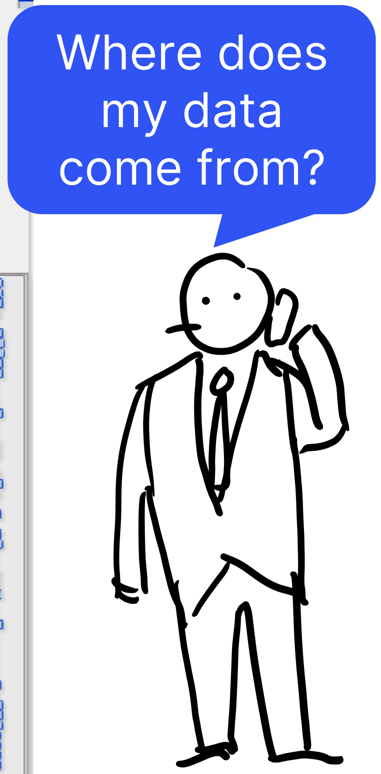

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Column level Usage Lineage

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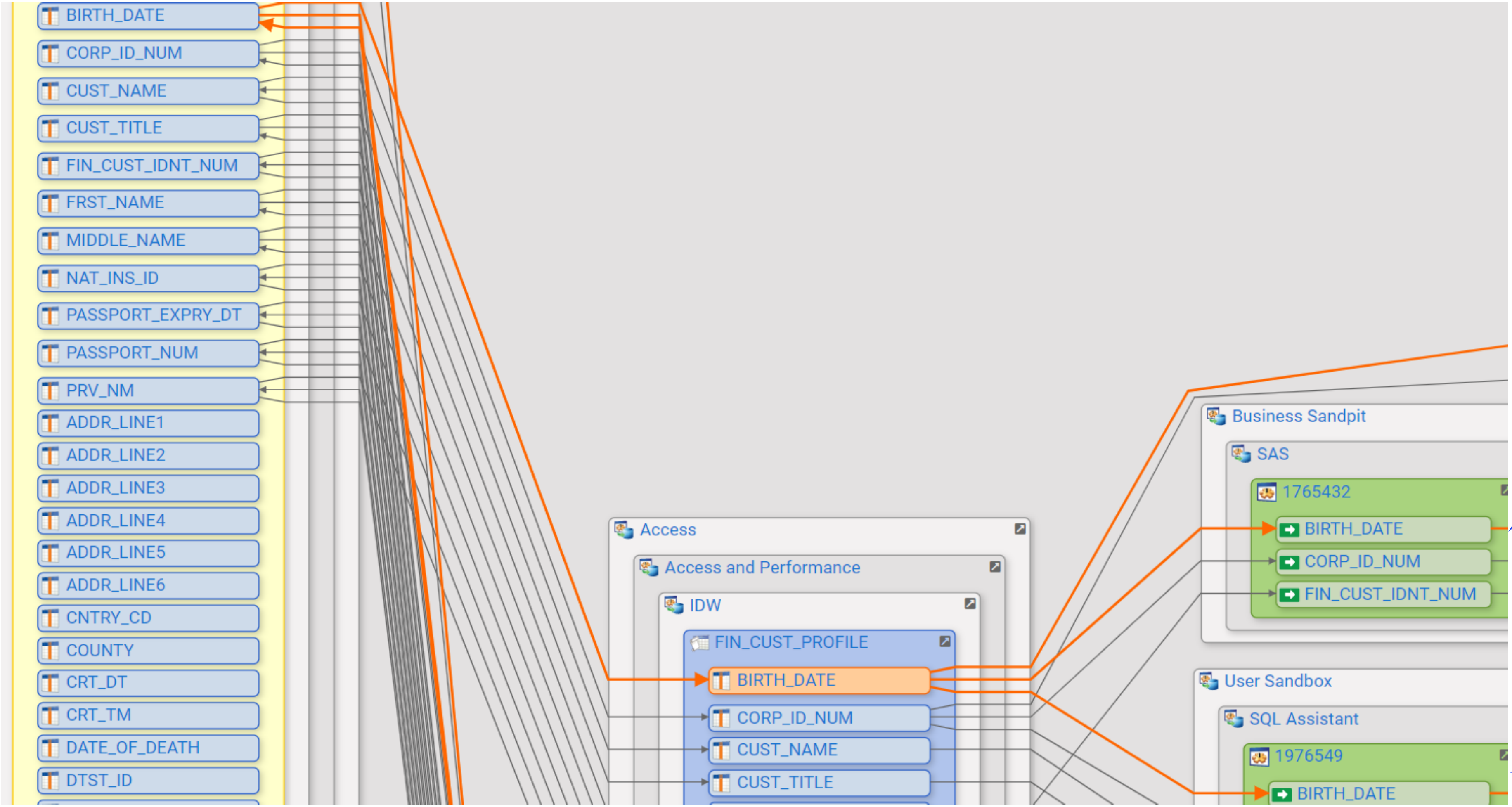

SAS Query

SQL Assistant Query

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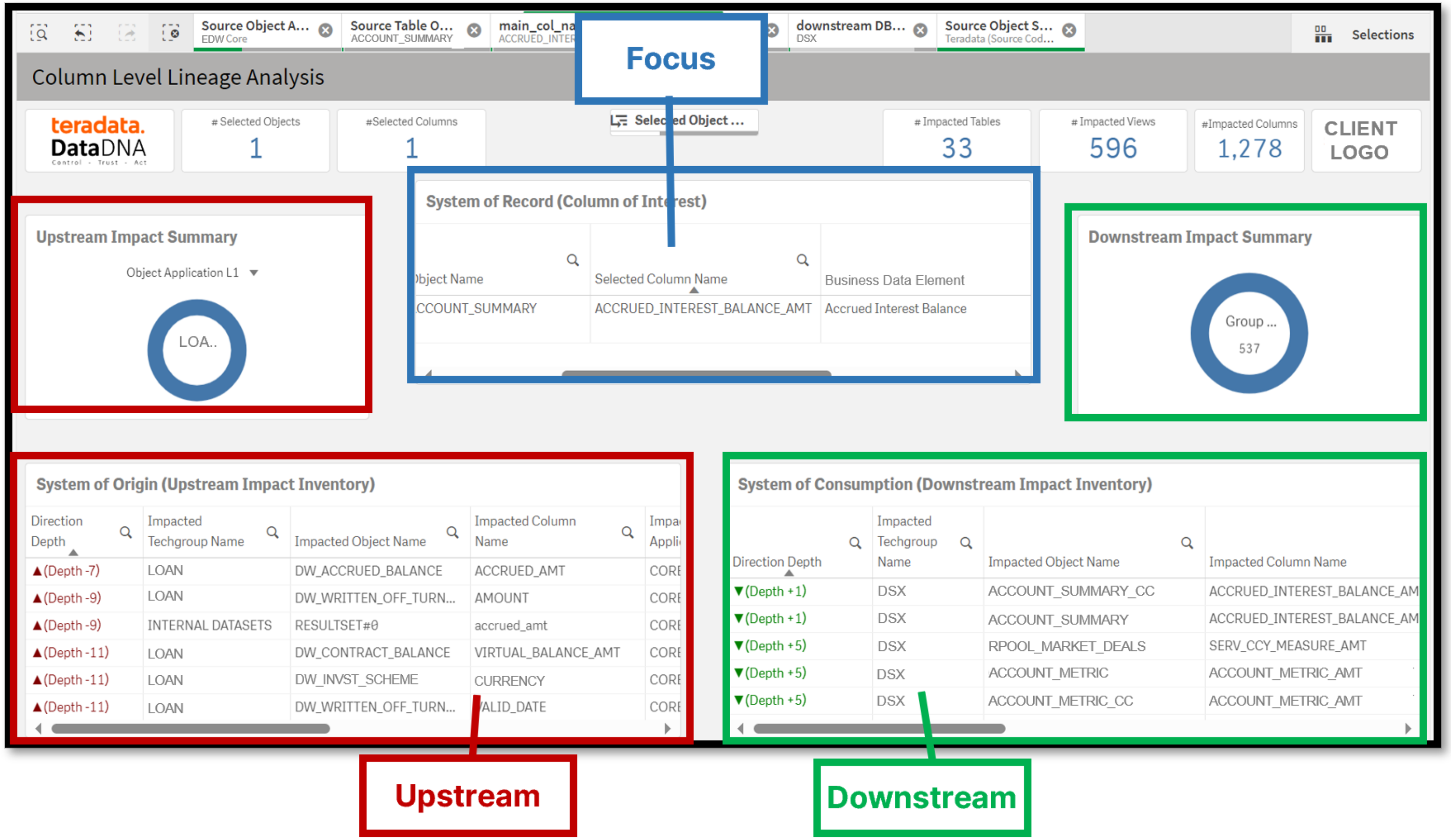
Lineage Analysis – Supports Data Lineage Navigation & Filtering

- ✓ Impact analysis simplifies navigation of data lineage in a complex data ecosystem
- ✓ It can be used for various analysis on top of fact-based data lineage.
- ✓ Hierarchies and filters to help finding relevant data assets, their dependencies and transformation rules.



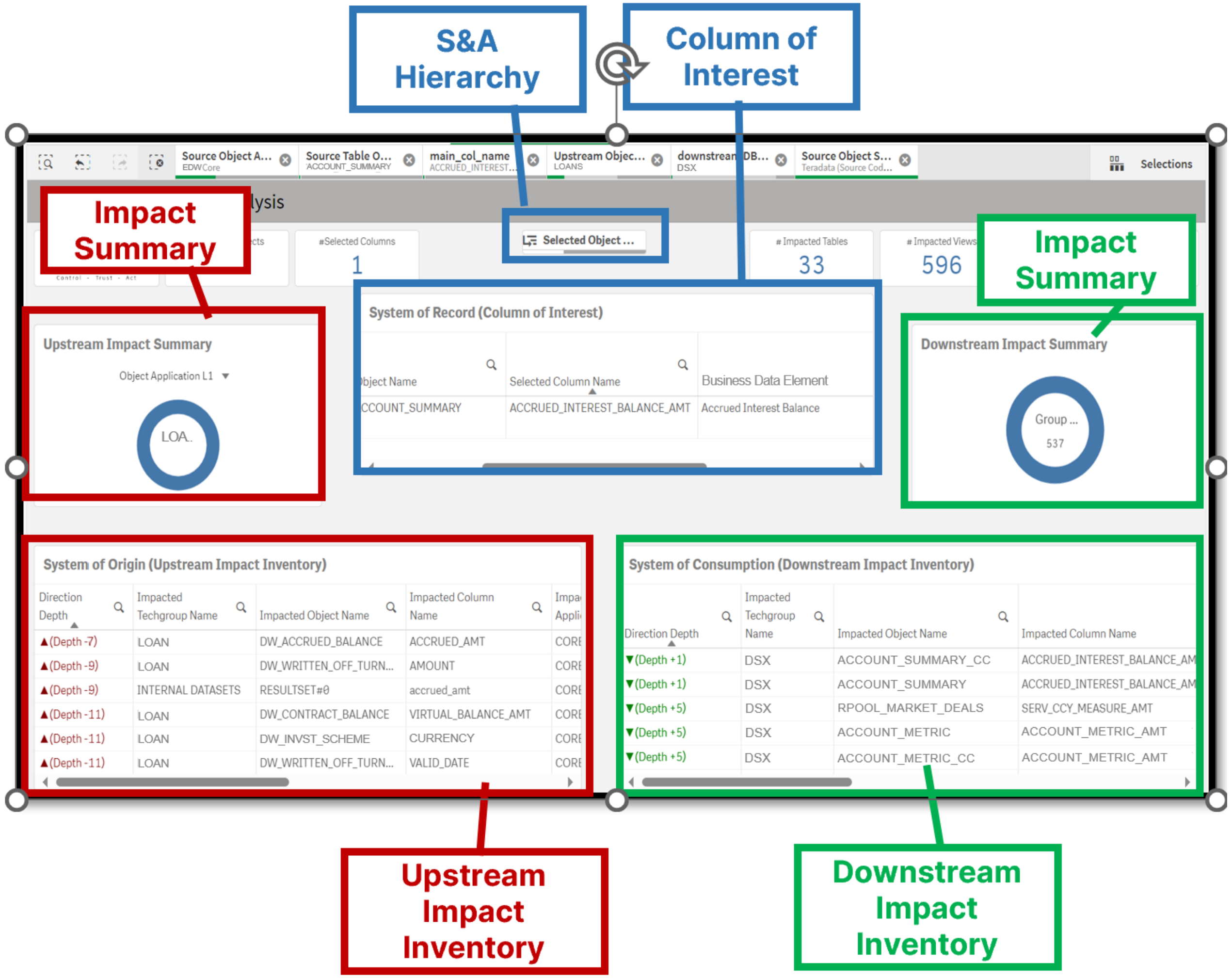
Column Level Lineage Analysis – Overview of Filters

General Filters – based on ingested metadata and hierarchies

- ✓ System & Application (S&A) Hierarchy
- ✓ Column of Interest
- ✓ Impact summaries
- ✓ Impact Inventory

Business Data Element (BDE) Filters - based on ingested BDE to EDW column mappings

- ✓ Used with Column of Interest



- ✓ Filters help to identify the Column of Interest and the change impacts Upstream & Downstream

Sheet Column Level Lineage Analysis

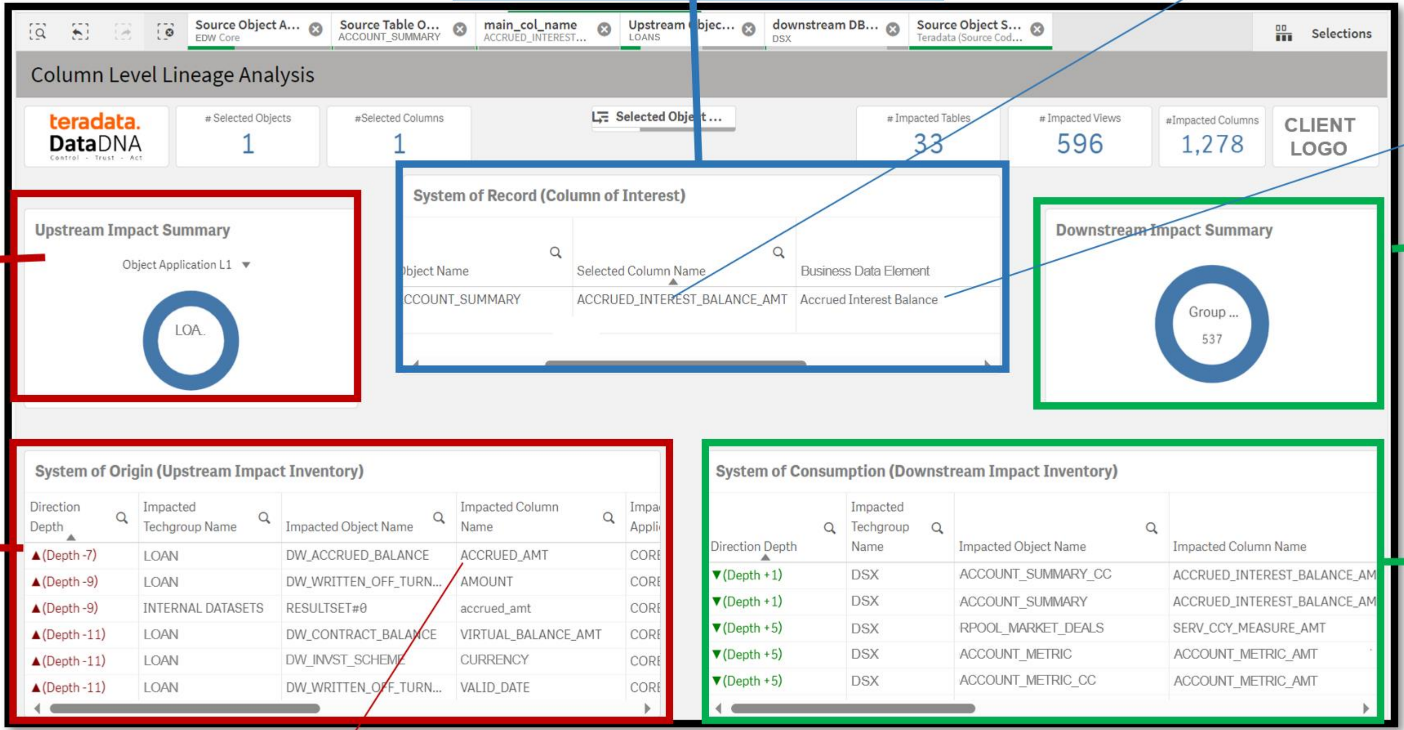
Column of Interest that is in the focus of the analysis

ACCOUNT_SUMMARY.ACCRUED_INTEREST_BALANCE_AMT

Filtering with BDEs is possible

Loans used as a filter for **upstream** flow

DSX used as a filter for **downstream** flow

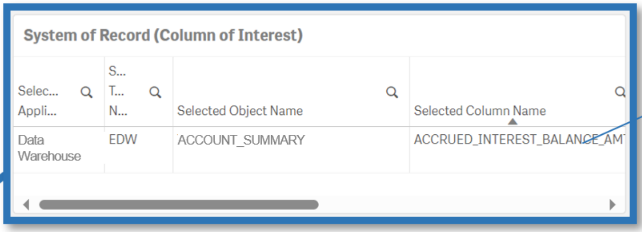


Upstream impacts from the Column of Interest

Downstream impacts from the Column of Interest

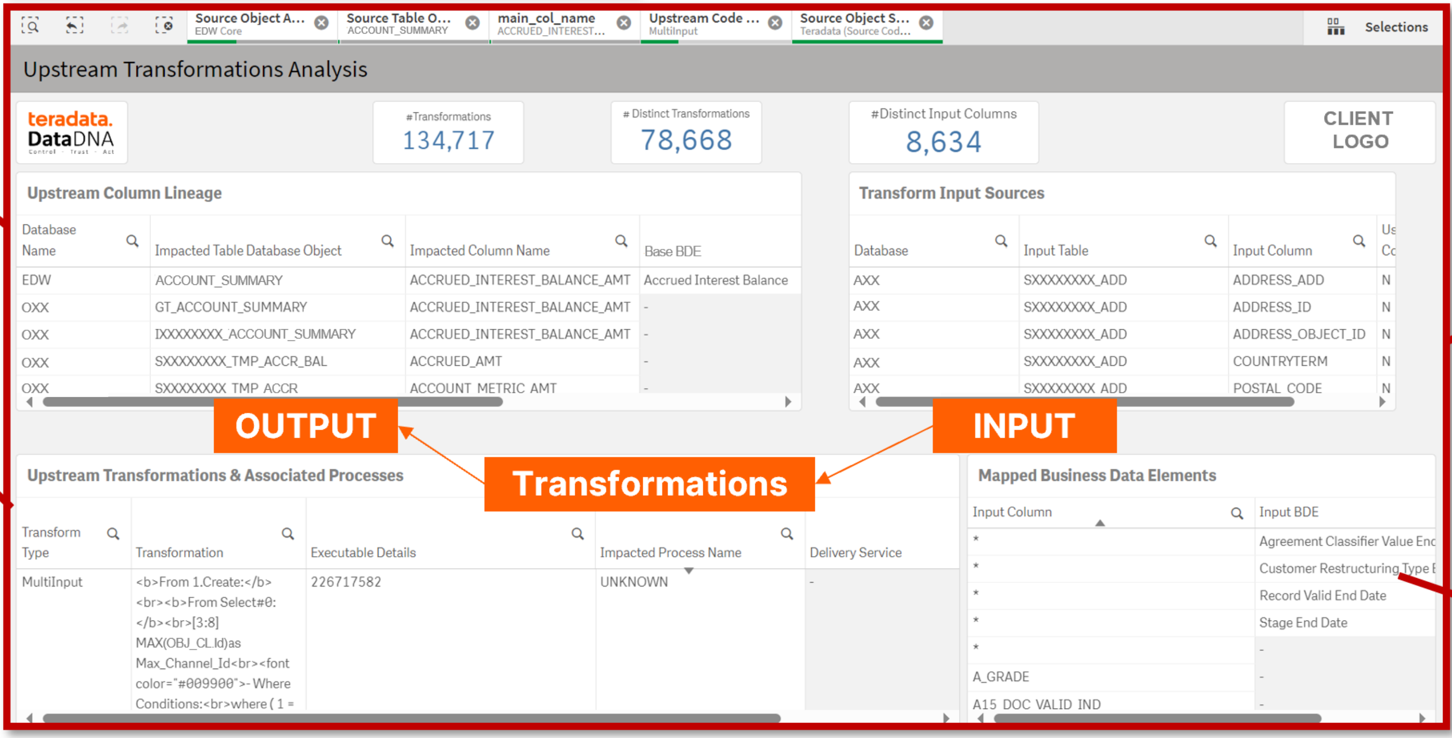
Loans DW_ACCRUED_BALANCE

Sheet Upstream Transformations Analysis



ACCOUNT_SUMMARY.ACCRUED_INTEREST_BALANCE_AMT

Object & Column of interest have been selected in previous sheets



Upstream Impacted objects & columns

All Upstream transformations for the selected column

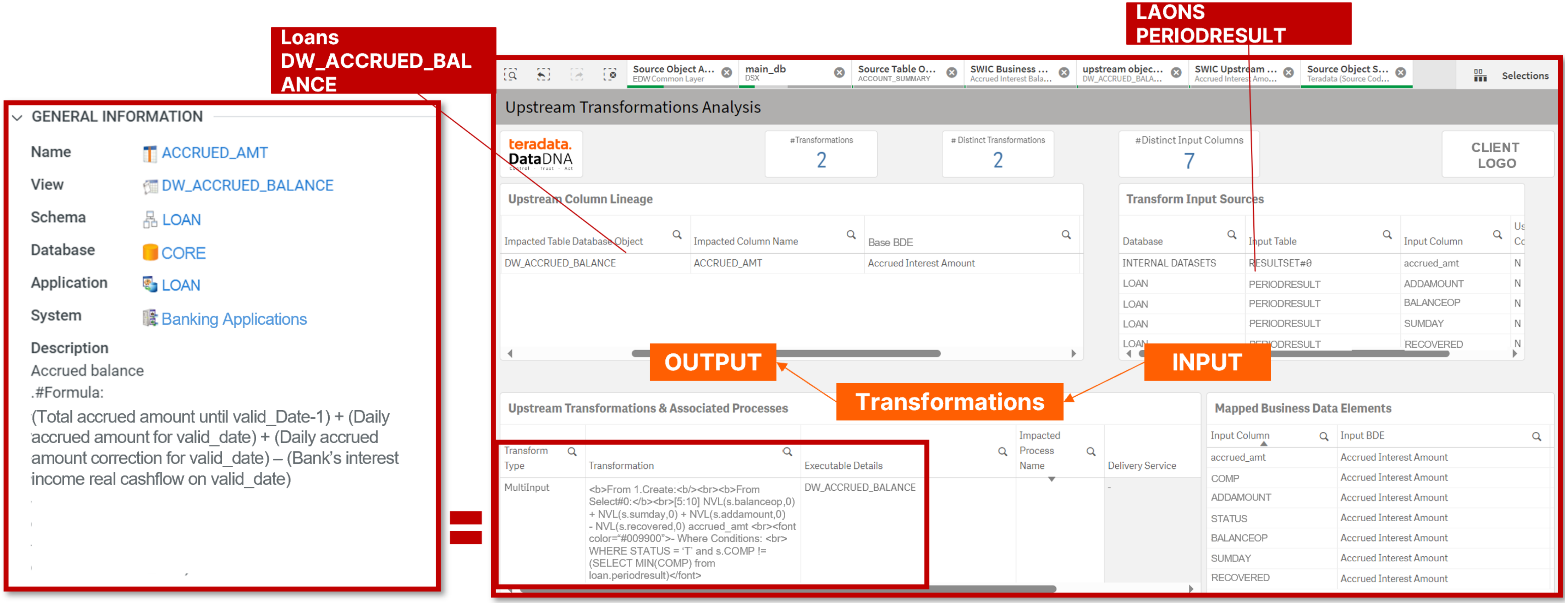
Helps to identify the hardest code blocks that are impacted

or where the main business rules are applied

Identifies data sources for the transformations

If selected column relates to BDE's they are listed here

Loans - DW_ACCRUED_BALANCE.Accrued_Amt



Analytics Dashboards

DataDNA Usage Analytics

Dashboard Overview

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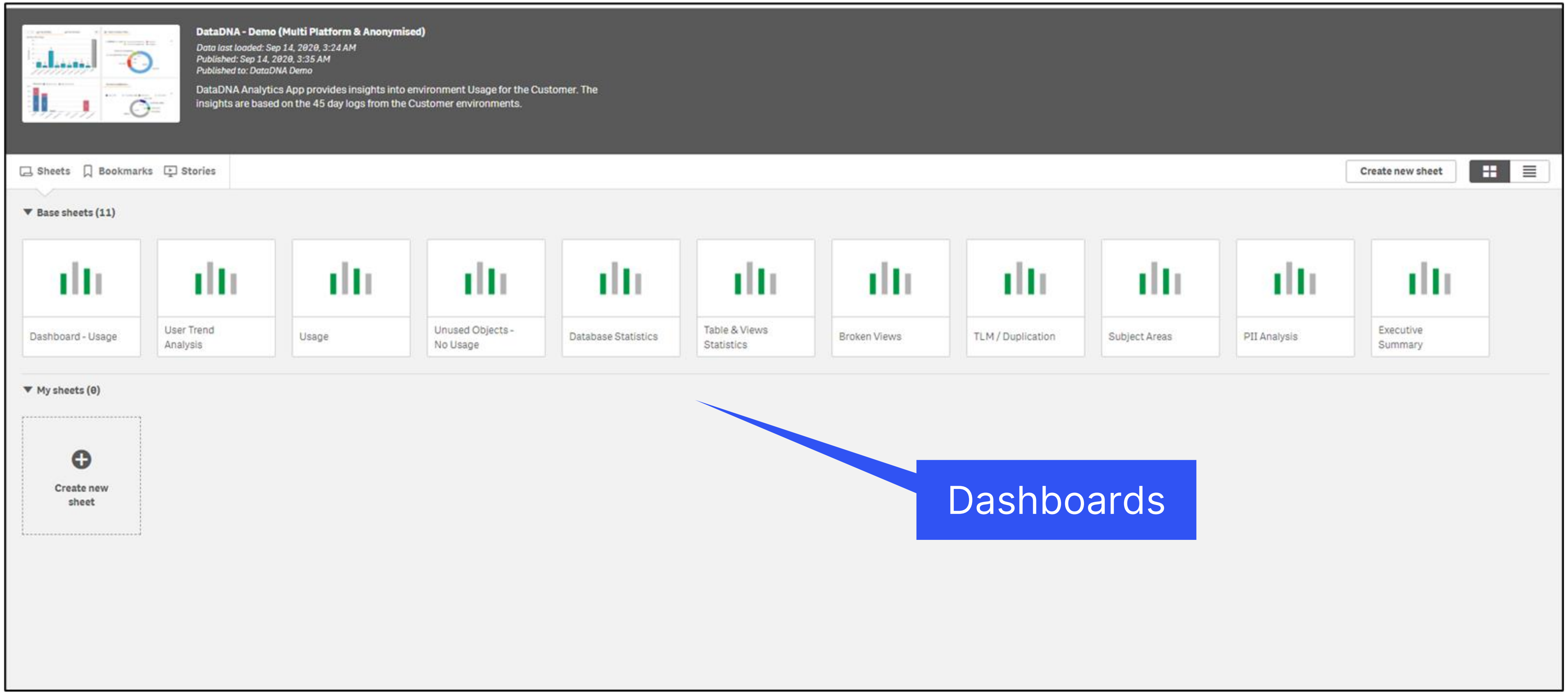

- ✓ Actionable Insight driven reports with visualizations
- ✓ Use Case driven Executive Summary & Actions

Usage Summary

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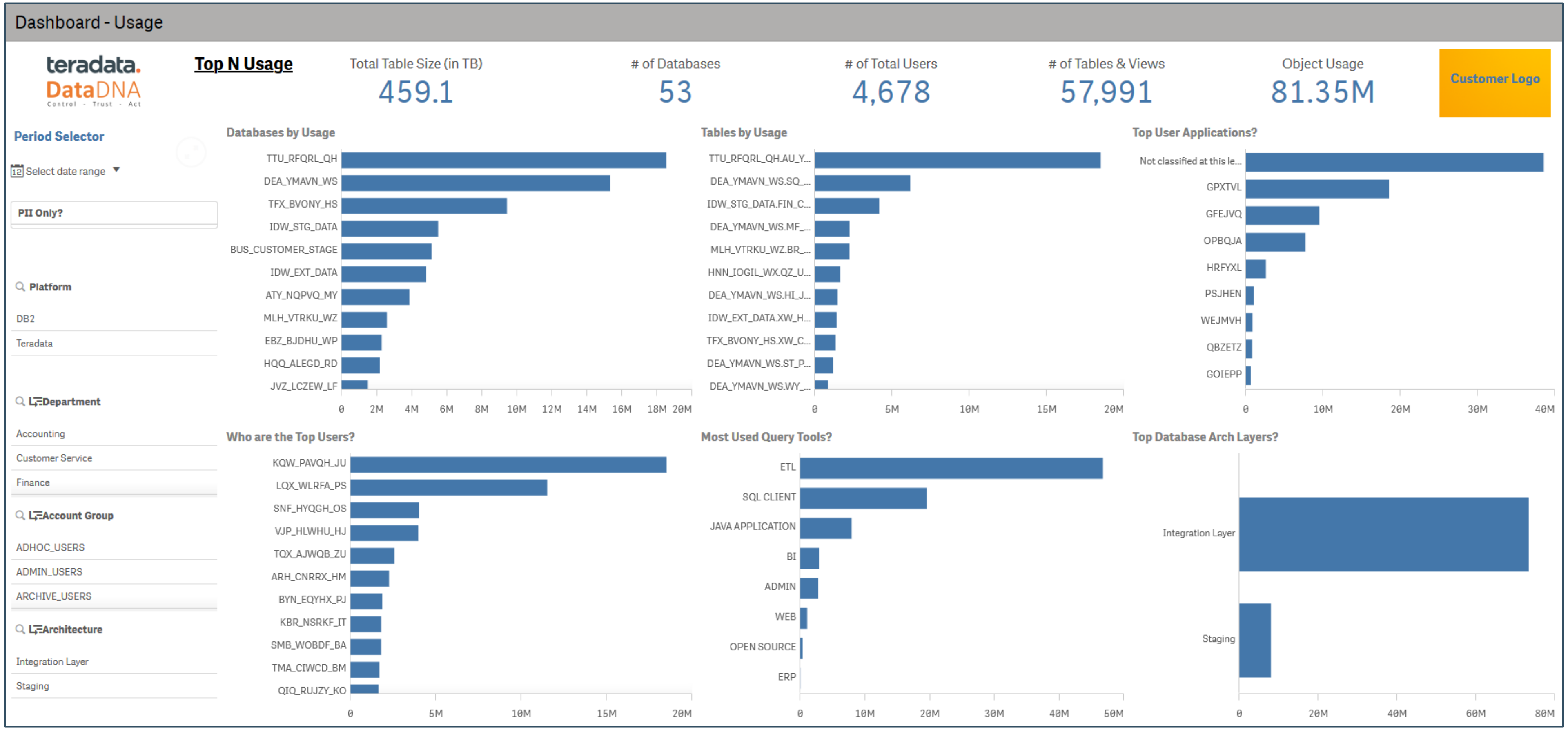

- ✓ Summarized report to inform Top N usage by Databases, Tables, Applications, Users, Architecture and Query Tools.
- ✓ Filters to drill into the details from the summary view. Additional filters available through 'Selections' pane.
- ✓ Filters selected & applied carry automatically to all Dashboard reports. Select filters from any report

User Trend Insights

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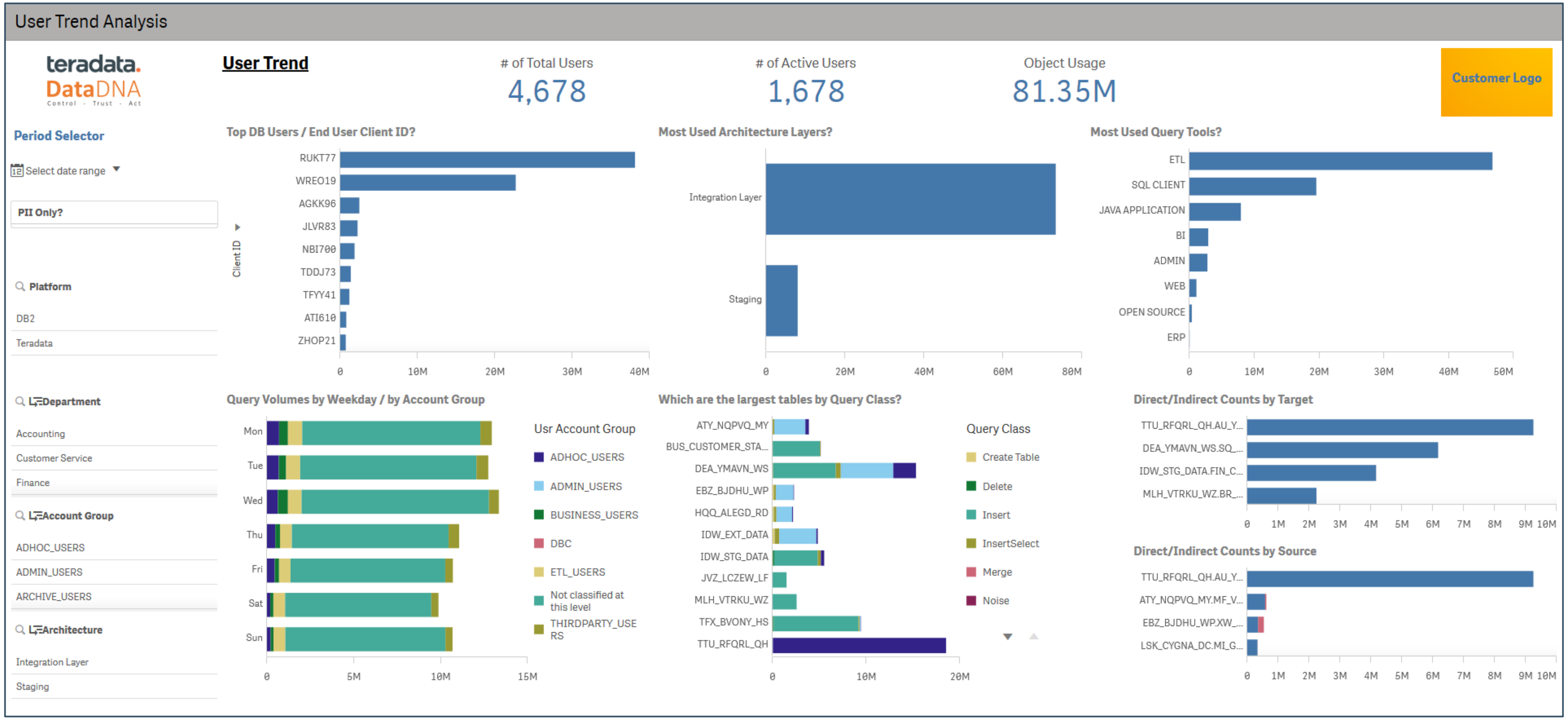

- ✓ Understand why & what users are doing, how are they accessing data and when are they accessing
- ✓ Types of queries executed through number of BI , ETL & Query tools.
- ✓ Indirect usage of the data through Views.
- ✓ Departmental/ Account Group behavior – requires departmental information to be supplied to enable this feature

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Data Product’s Data Domain Analyss

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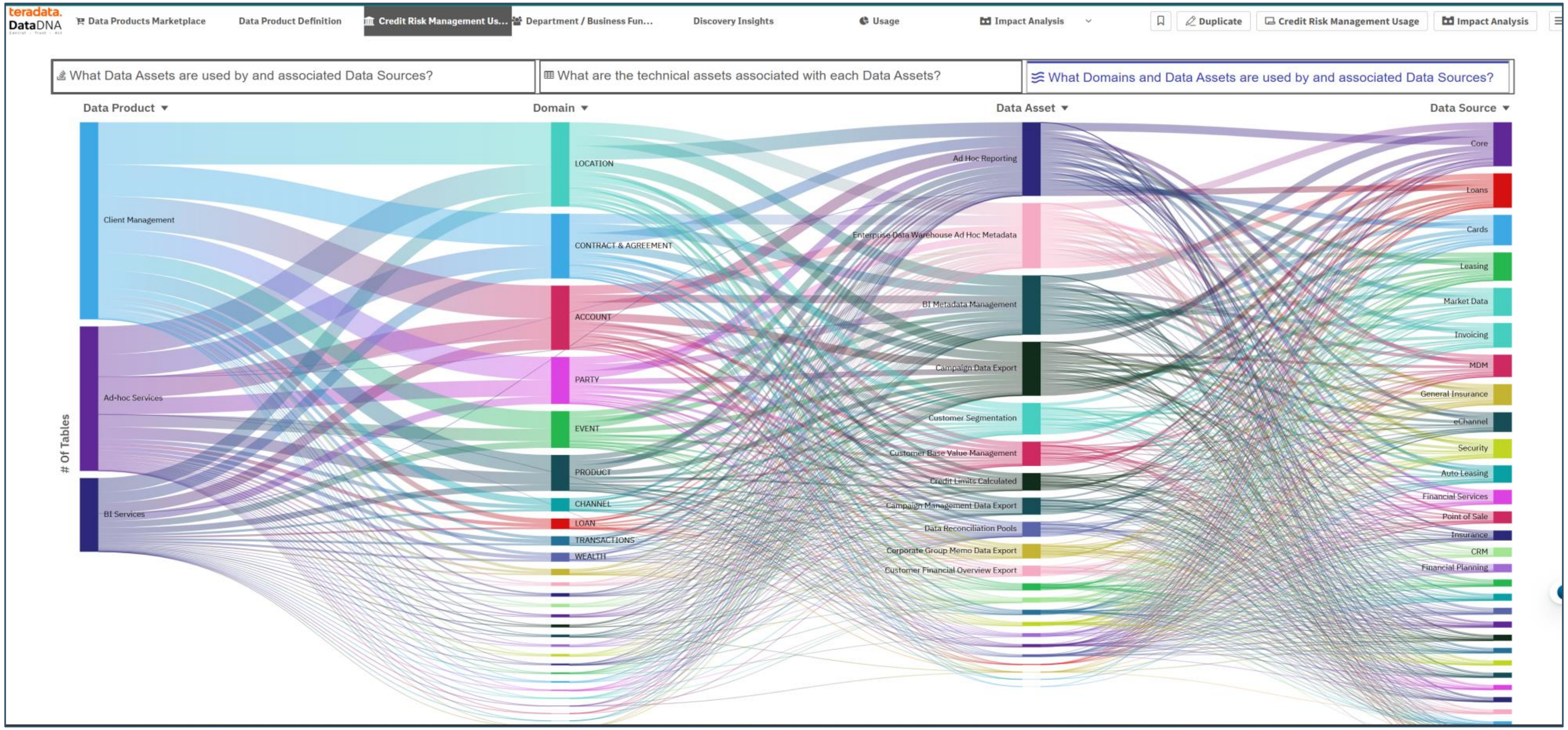

- ✓ Data Products and Data Domains mapping to Sources systems
- ✓ Data Domain and Data Assets mapped to physical columns
- ✓ Column level lineage driven impact analyses deployed to provide map Data Assets to their ultimate source systems.

Data Product's Usage Analyss

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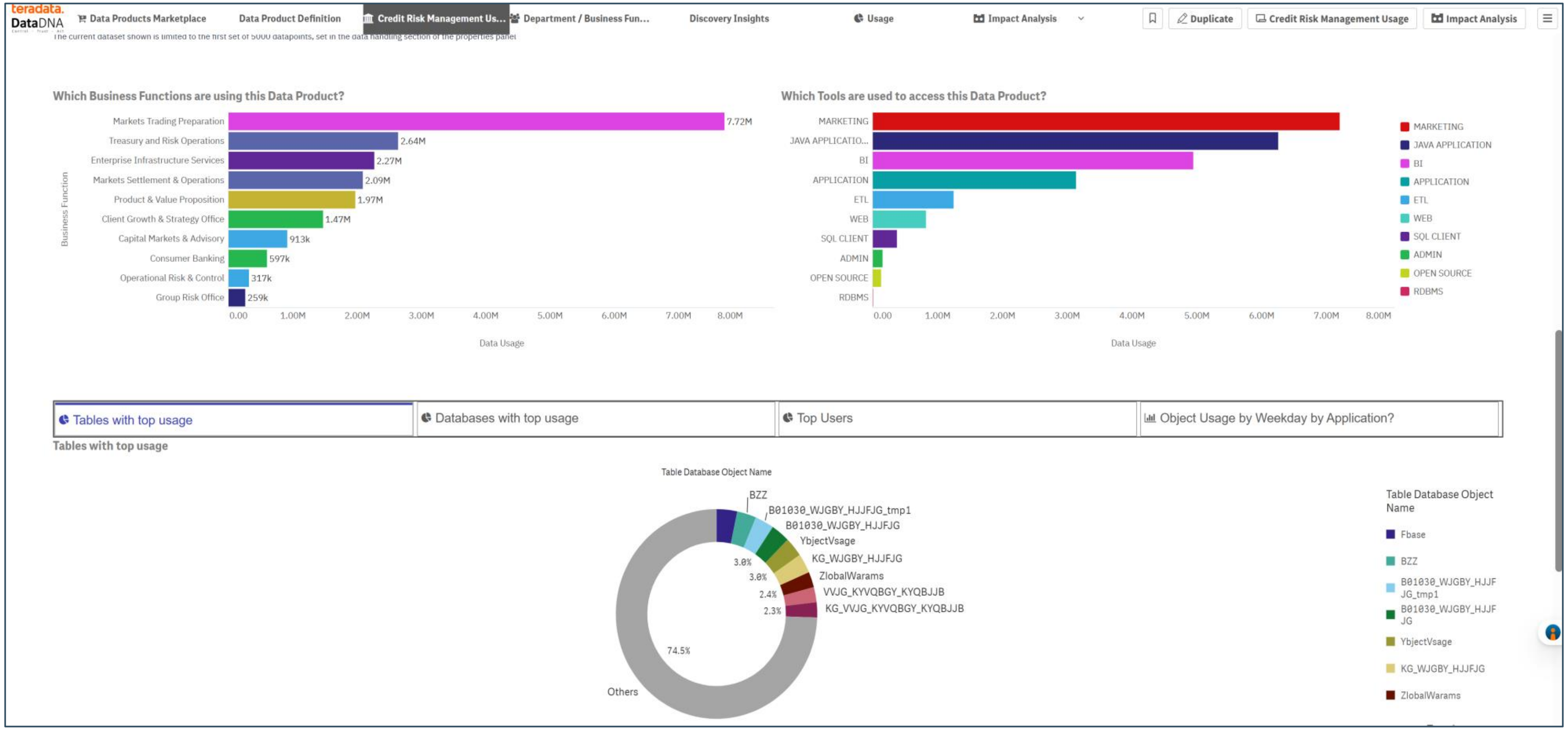

- ✓ Business Function usage of Data Products
- ✓ Query tools access of Data Products
- ✓ Data Products tables analysis

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Used / Unused Data Summary

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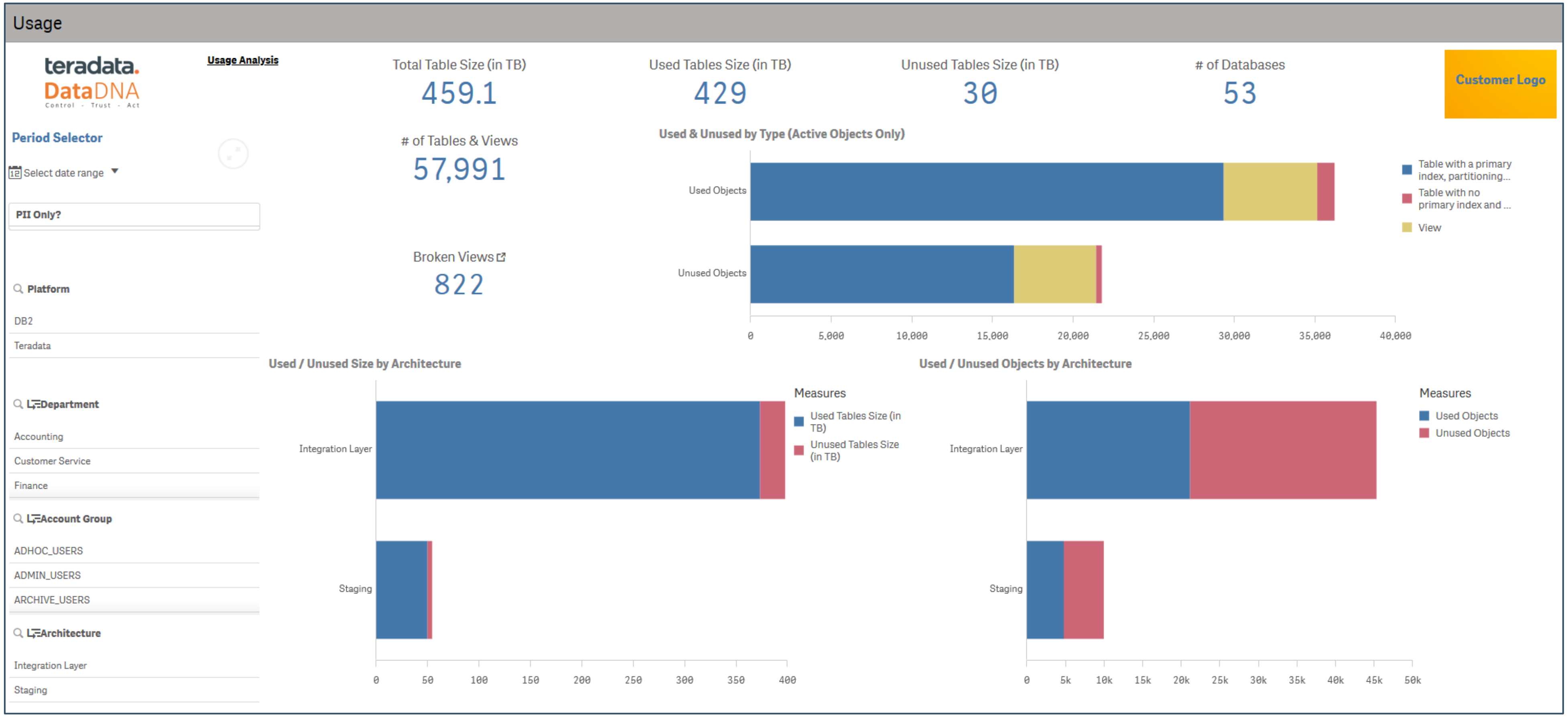

- ✓ For the ecosystem, informs summary of used and unused data (objects) for the analysis period in scope.
- ✓ Periodic refresh can inform this over a time period i.e. timeseries analysis
- ✓ Provides ability to drill down into architecture & databases that have objects that are unused.

Unused Data - Details

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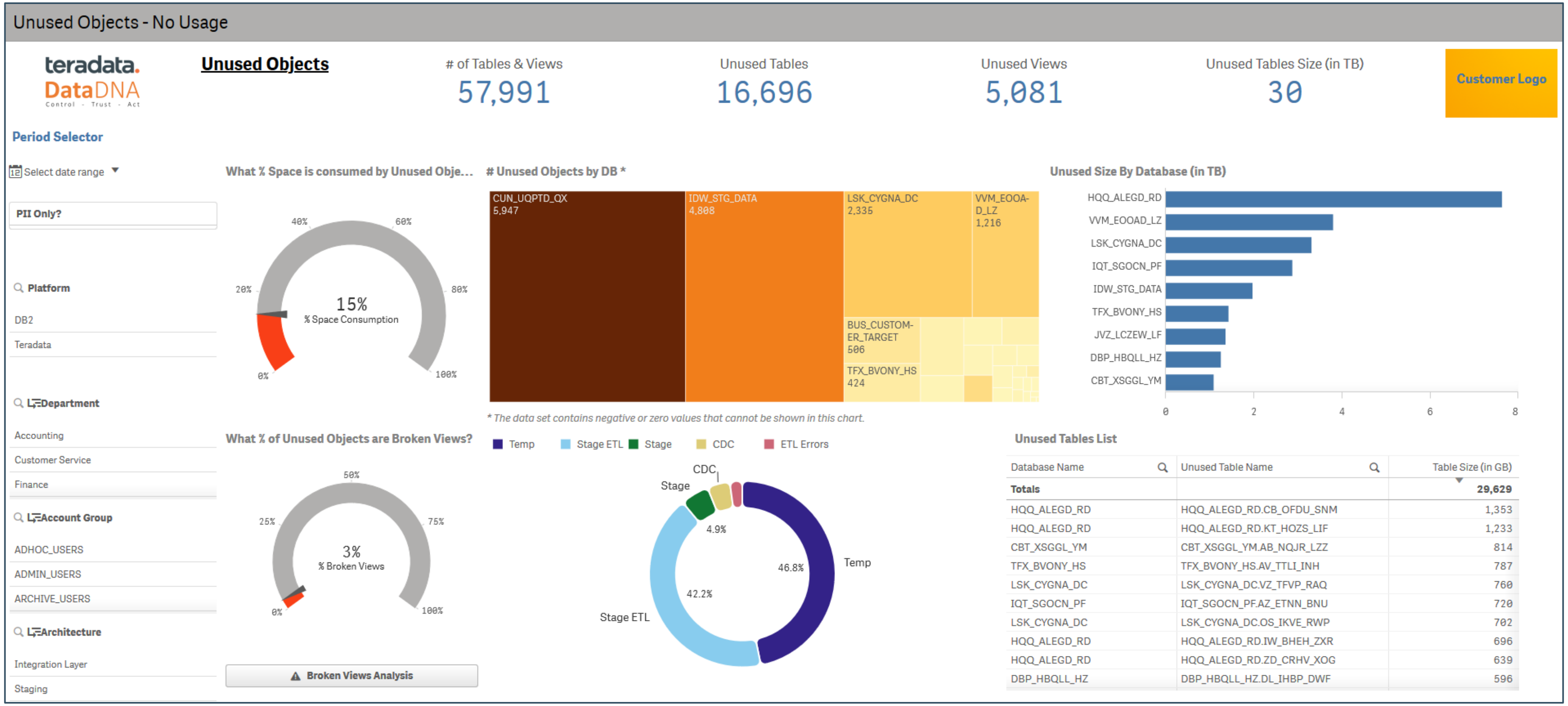

Database Name	Unused Table Name	Table Size (in GB)
Totals		29,629
HQQ_ALEGD_RD	HQQ_ALEGD_RD.CB_OFDU_SNM	1,353
HQQ_ALEGD_RD	HQQ_ALEGD_RD.KT_HOZS_LIF	1,233
CBT_XSGGL_YM	CBT_XSGGL_YM.AB_NQJR_LZZ	814
TFX_BVONY_HS	TFX_BVONY_HS.AV_TTLI_INH	787
LSK_CYGNA_DC	LSK_CYGNA_DC.VZ_TFVP_RAQ	760
IQT_SGOCN_PF	IQT_SGOCN_PF.AZ_ETNN_BNU	720
LSK_CYGNA_DC	LSK_CYGNA_DC.OS_IKVE_RWP	702
HQQ_ALEGD_RD	HQQ_ALEGD_RD.IW_BHEH_ZXR	696
HQQ_ALEGD_RD	HQQ_ALEGD_RD.ZD_CRHV_XOG	639
DBP_HBQLL_HZ	DBP_HBQLL_HZ.DL_IHBP_DWF	596

- ✓ Informs % of space consumed by the unused objects – candidates for clean up / archive
- ✓ Analyze unused objects by Databases and drill down to actual object list. More details on objects can be further analyzed through 'Object Statistics' (covered here)
- ✓ Leads into Broken views analysis / insights if there are any in the databases.

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Database Statistics

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Database Statistics

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Database Statistics

Total Table Size (in TB)	Space Available (in TB)	Max DB Space (in TB)	# of Databases	# of Tables & Views
459.1	106	568	53	57,991

Customer Logo

Period Selector

Select date range

PII Only?

Platform
- DB2
- Teradata

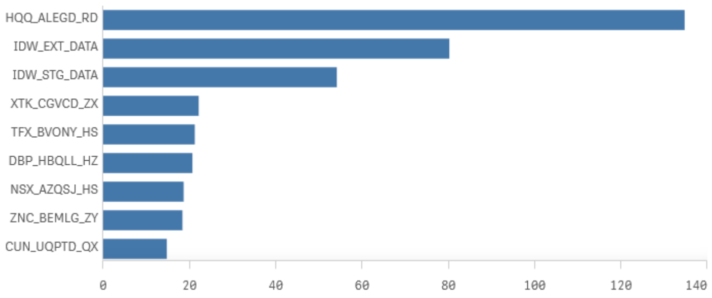
Department
- Accounting
- Customer Service
- Finance

Account Group
- ADHOC_USERS
- ADMIN_USERS
- ARCHIVE_USERS

Architecture
- Integration Layer
- Staging

Total Database Objects by Architecture Layer

Which are the largest databases by size (in TB)?



Database Metric (Size in GB)

Database Name	Total Current PERM	Total Peak PERM	Peak PERM Skew Factor	Max PERM Space Size	Impact Peak	Impact PERM	Max Current PERM	Max Peak PERM	Min Current PERM	Min Peak PERM
Totals	**473,079**	**476,266**	**3,000**	**288**	**497,388**	**493,384**	**245**	**247**	**232**	**233**
HQQ_ALEGD_RD	137,973	137,974	100	74	143,193	143,192	71	71	68	68
IDW_EXT_DATA	82,144	82,144	100	46	82,479	82,479	41	41	41	41
IDW_STG_DATA	55,435	56,405	100	33	58,318	57,349	28	29	27	28
XTK_CGVCD_ZX	22,654	22,654	100	13	22,861	22,861	11	11	11	11
TFX_BVONY_HS	21,725	21,800	100	15	24,341	24,260	12	12	10	10
DBP_HBQLL_HZ	21,160	21,160	100	13	21,549	21,549	11	11	10	10
NSX_AZQSJ_HS	19,090	19,342	100	12	19,656	19,408	10	10	9	9
ZNC_BEMLG_ZY	18,802	18,802	100	11	19,262	19,262	10	10	9	9
CUN_UQPTD_QX	15,094	15,830	100	12	20,028	19,301	10	10	7	8
MLH_VTRKU_WZ	12,748	12,748	100	6	12,820	12,820	6	6	6	6
VVM_EOOAD_LZ	9,667	9,817	100	7	11,215	9,728	5	6	5	5
IQT_SGOCN_PF	8,024	8,030	100	4	8,194	8,189	4	4	4	4

- ✓ Provides dictionary level statistics about the databases on the platform.
- ✓ Informs largest databases by size on the platform.

Object Statistics

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Table & Views Statistics

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Table & Views Statistics

Total Table Size (in TB)	Space Available (in TB)	Max DB Space (in TB)	# of Databases	# of Tables & Views
459	106	568	53	57,991

Customer Logo

Period Selector

Select date range

PII Only?

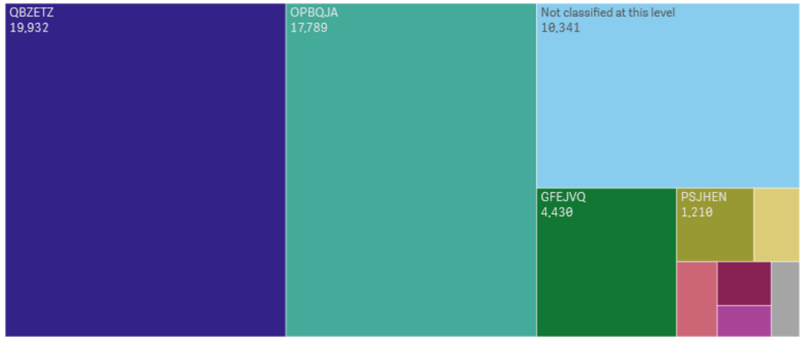
Platform
- DB2
- Teradata

Department
- Accounting
- Customer Service
- Finance

Account Group
- ADHOC_USERS
- ADMIN_USERS
- ARCHIVE_USERS

Architecture
- Integration Layer
- Staging

Table Count by Application



Which are the largest tables by size (in TB)?

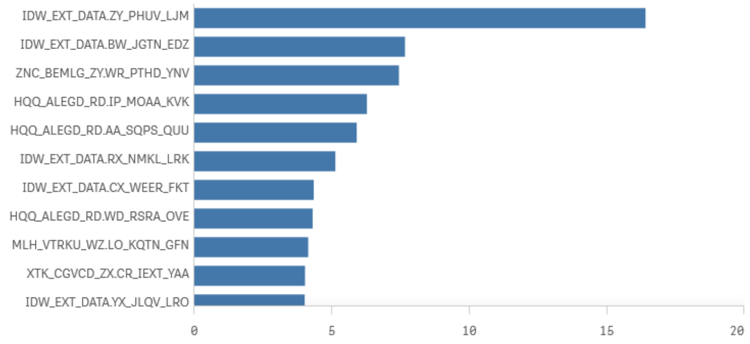


Table Metric (Size in GB)

Table Name	Current PERM Size	Direct Count	Indirect Count	LastAccessDate	DateCreated	Table Database Object Kind	Sourc
Totals	**470,155**	**59,371,790**	**21,977,119**				
IDW_EXT_DATA.ZY_PHUV_LJM	16,801	446	8,416	2020-04-02	2019-09-06	Table with a primary index, partitioning, or both	Show
IDW_EXT_DATA.BW_JGTN_EDZ	7,842	505	407,486	2020-04-02	2018-08-03	Table with a primary index, partitioning, or both	Show
ZNC_BEMLG_ZY.WR_PTHD_YNV	7,618	4,074	12,072	2020-04-02	2019-08-22	Table with a primary index, partitioning, or both	Show
HQQ_ALEGD_RD.IP_MOAA_KVK	6,424	31	18,154	2020-04-02	2017-09-16	Table with a primary index, partitioning, or both	Show
HQQ_ALEGD_RD.AA_SQPS_QUU	6,045	13	658	2020-04-02	2017-05-26	Table with a primary index, partitioning, or both	Show
IDW_EXT_DATA.RX_NMKL_LRK	5,249	32	1,788	2020-04-02	2017-11-17	Table with a primary index, partitioning, or both	Show
IDW_EXT_DATA.CX_WEER_FKT	4,444	1,458	11,207	2020-04-02	2018-01-04	Table with a primary index, partitioning, or both	Show
HQQ_ALEGD_RD.WD_RSRA_OVE	4,405	306	46,990	2020-04-02	2018-03-04	Table with a primary index, partitioning, or both	Show
MLH_VTRKU_WZ.LO_KQTN_GFN	4,238	165	5,210	2020-04-02	2018-06-28	Table with a primary index, partitioning, or both	Show
XTK_CGVCD_ZX.CR_IEXT_YAA	4,118	310	2,622	2020-04-02	2019-11-14	Table with a primary index, partitioning, or both	Show

- ✓ Provides dictionary level statistics about the tables across the databases.
- ✓ Informs largest tables by size on the platform.
- ✓ Provides ability to show / navigate to the data lineage of a particular object of interest.

Broken Views

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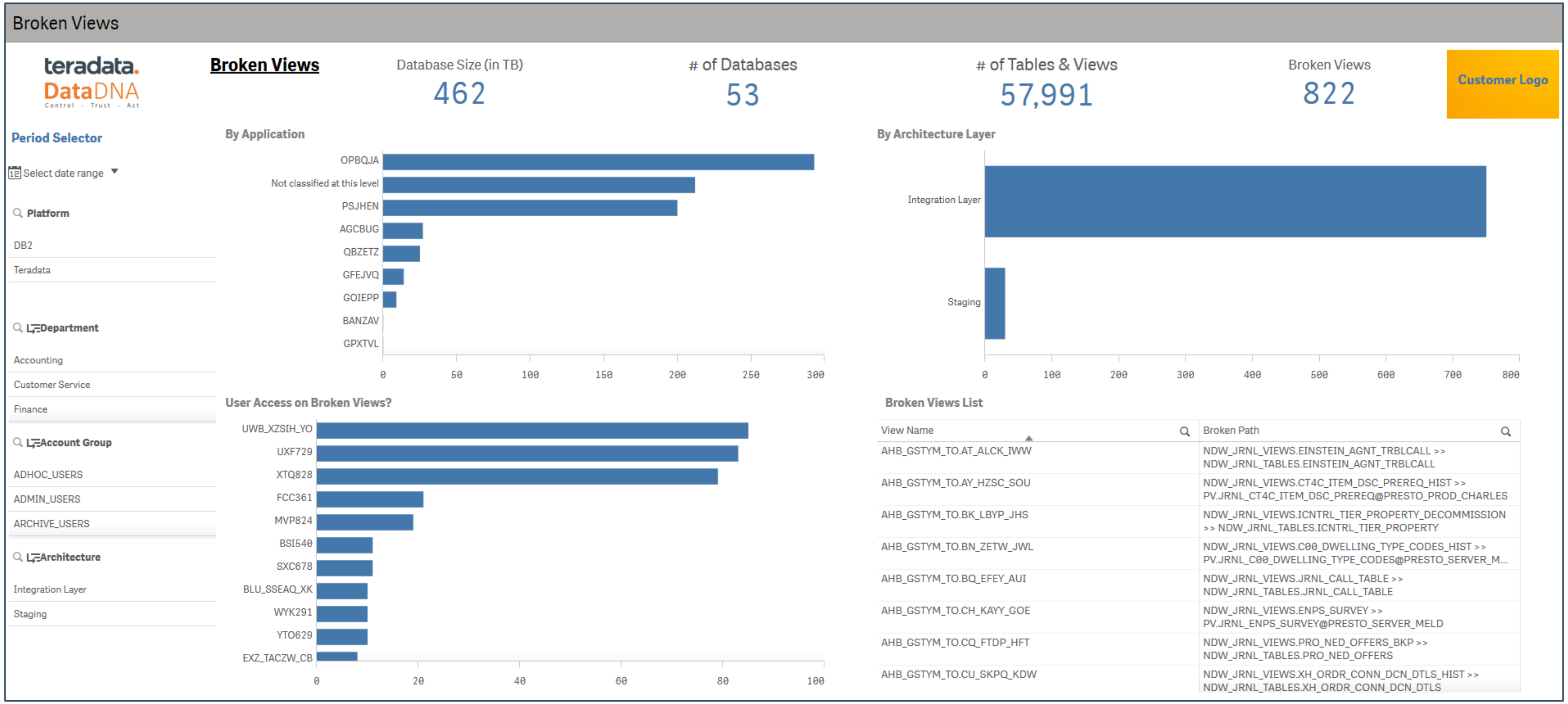

- ✓ Summary statistics on number of Broken views by Architecture and Application.
- ✓ 'Who' has attempted to access these views during the analysis period.
- ✓ Detailed list of broken views and informs the broken view hierarchy i.e. broken / missing object that caused the view to break.

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Fact based lineage enables cross platform duplication identification

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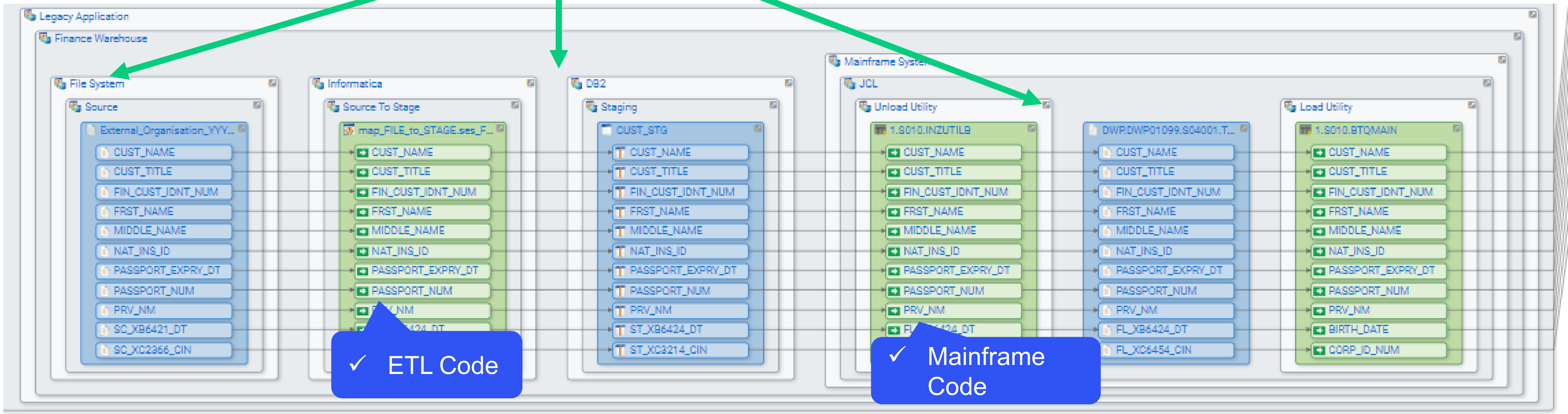
✓ Duplicated Tables and files created by code or user queries



✓ ETL Code

✓ Mainframe Code

- ✓ Fact based lineage across environments – tells us how data moves from one dataset to another
- ✓ This enables DataDNA to automatically identify where data is being duplicated across one platform or a group of platforms
- ✓ Leveraging the usage analytics within DataDNA, customers can identify which datasets are being used (or not) and by which users/departments/tools

Duplicate Set Analysis

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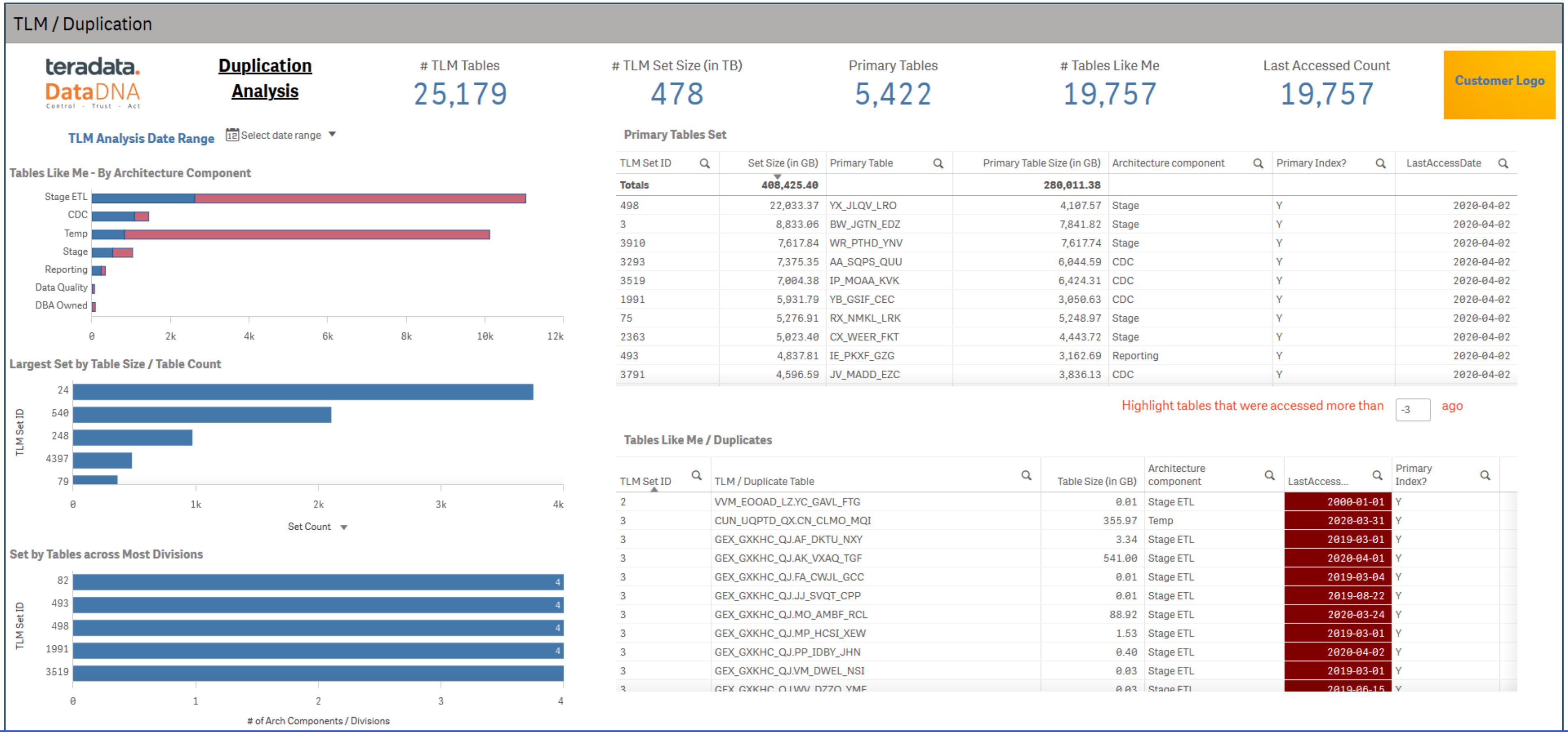

Primary Tables Set

TLM Set ID	Set Size (in GB)	Primary Table	Primary Table Size (in GB)	Architecture component	Primary Index?	LastAccessDate
Totals	408,425.40		280,011.38			
498	22,033.37	YX_JLQV_LRO	4,107.57	Stage	Y	2020-04-02
3	8,833.06	BW_JGTN_EDZ	7,841.82	Stage	Y	2020-04-02
3910	7,617.84	WR_PTHD_YNV	7,617.74	Stage	Y	2020-04-02
3293	7,375.35	AA_SQPS_QUU	6,044.59	CDC	Y	2020-04-02
3519	7,004.38	IP_MOAA_KVK	6,424.31	CDC	Y	2020-04-02
1991	5,931.79	YB_GSIF_CEC	3,050.63	CDC	Y	2020-04-02
75	5,276.91	RX_NMKL_LRK	5,248.97	Stage	Y	2020-04-02
2363	5,023.40	CX_WEER_FKT	4,443.72	Stage	Y	2020-04-02
493	4,837.81	IE_PKXF_GZG	3,162.69	Reporting	Y	2020-04-02
3791	4,596.59	JV_MADD_EZC	3,836.13	CDC	Y	2020-04-02

Highlight tables that were accessed more than -3 ago

Tables Like Me / Duplicates

TLM Set ID	TLM / Duplicate Table	Table Size (in GB)	Architecture component	LastAccess...	Primary Index?
2	VVM_EOOAD_LZ.YC_GAVL_FTG	0.01	Stage ETL	2000-01-01	Y
3	CUN_UQPTD_QX.CN_CLMO_MQI	355.97	Temp	2020-03-31	Y
3	GEX_GXKHC_QJ.AF_DKTU_NXY	3.34	Stage ETL	2019-03-01	Y
3	GEX_GXKHC_QJ.AK_VXAQ_TGF	541.00	Stage ETL	2020-04-01	Y
3	GEX_GXKHC_QJ.FA_CWJL_GCC	0.01	Stage ETL	2019-03-04	Y
3	GEX_GXKHC_QJ.JJ_SVQT_CPP	0.01	Stage ETL	2019-08-22	Y
3	GEX_GXKHC_QJ.MO_AMBF_RCL	88.92	Stage ETL	2020-03-24	Y
3	GEX_GXKHC_QJ.MP_HCSI_XEW	1.53	Stage ETL	2019-03-01	Y
3	GEX_GXKHC_QJ.PP_IDBY_JHN	0.40	Stage ETL	2020-04-02	Y
3	GEX_GXKHC_QJ.VM_DWEL_NSI	0.03	Stage ETL	2019-03-01	Y
3	GEX_GXKHC_QJ.WV_DZZQ_YME	0.03	Stage ETL	2019-06-15	Y

- ✓ Comprehensive analysis on data duplication i.e. tables are same or similar in structures by Architecture and/or Application.
- ✓ The information is grouped as ‘Sets’ that represent Primary tables and associated ‘Tables Like Me’ (TLM).
- ✓ This allows analysis through targeting top ‘Sets’ by Size or # of Tables.
- ✓ Informs ‘when’ the tables were last accessed which can facilitate archiving / clean up decisions.

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Subject Area Insights

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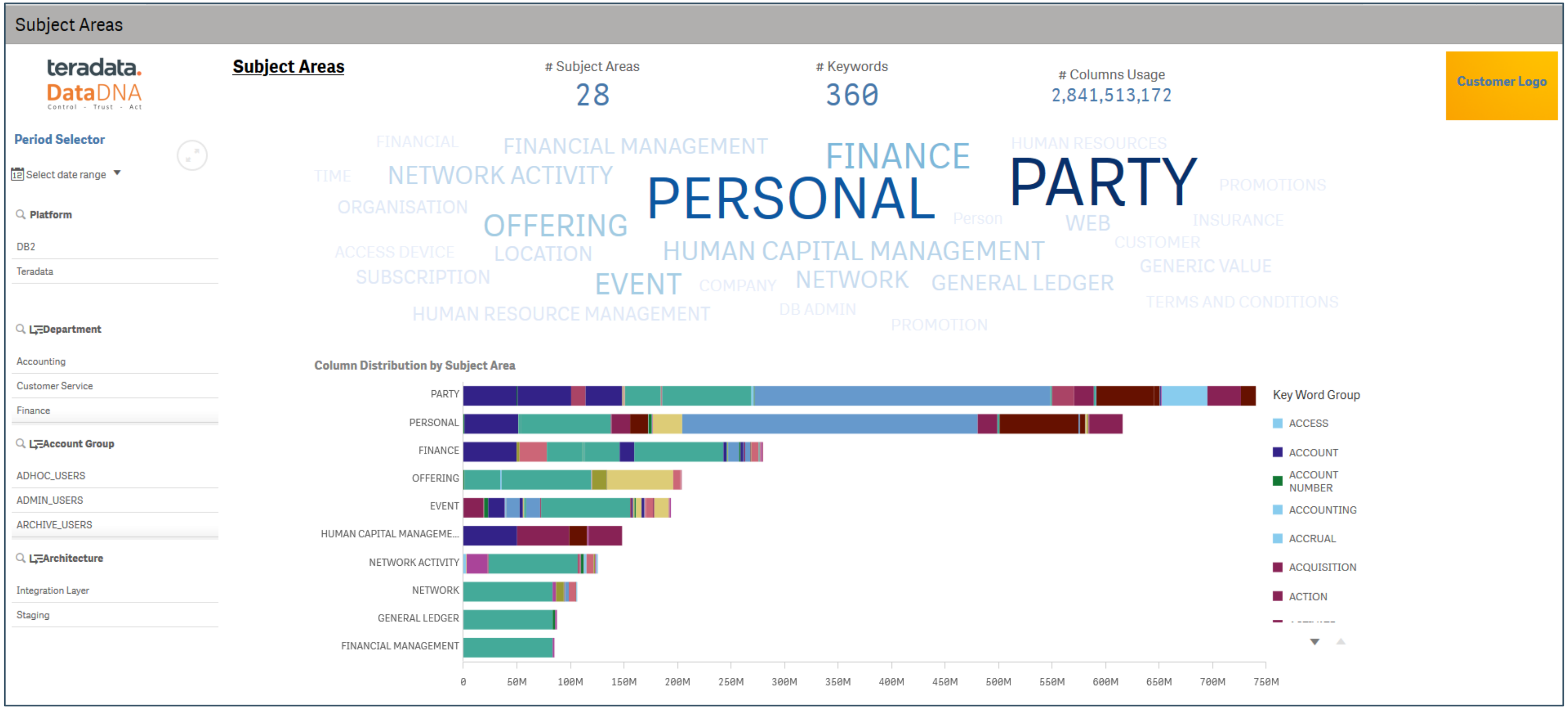

- ✓ Subject Area/Domain analysis that is powered by very mature inference engine driven by column usage.
- ✓ Quickly analyze the most common / popular subject areas with ability to drill down to underlying applications, tables and user activity
- ✓ Ability to inform / lead migration use cases e.g. moving to cloud, data mart consolidations.

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PII Insights

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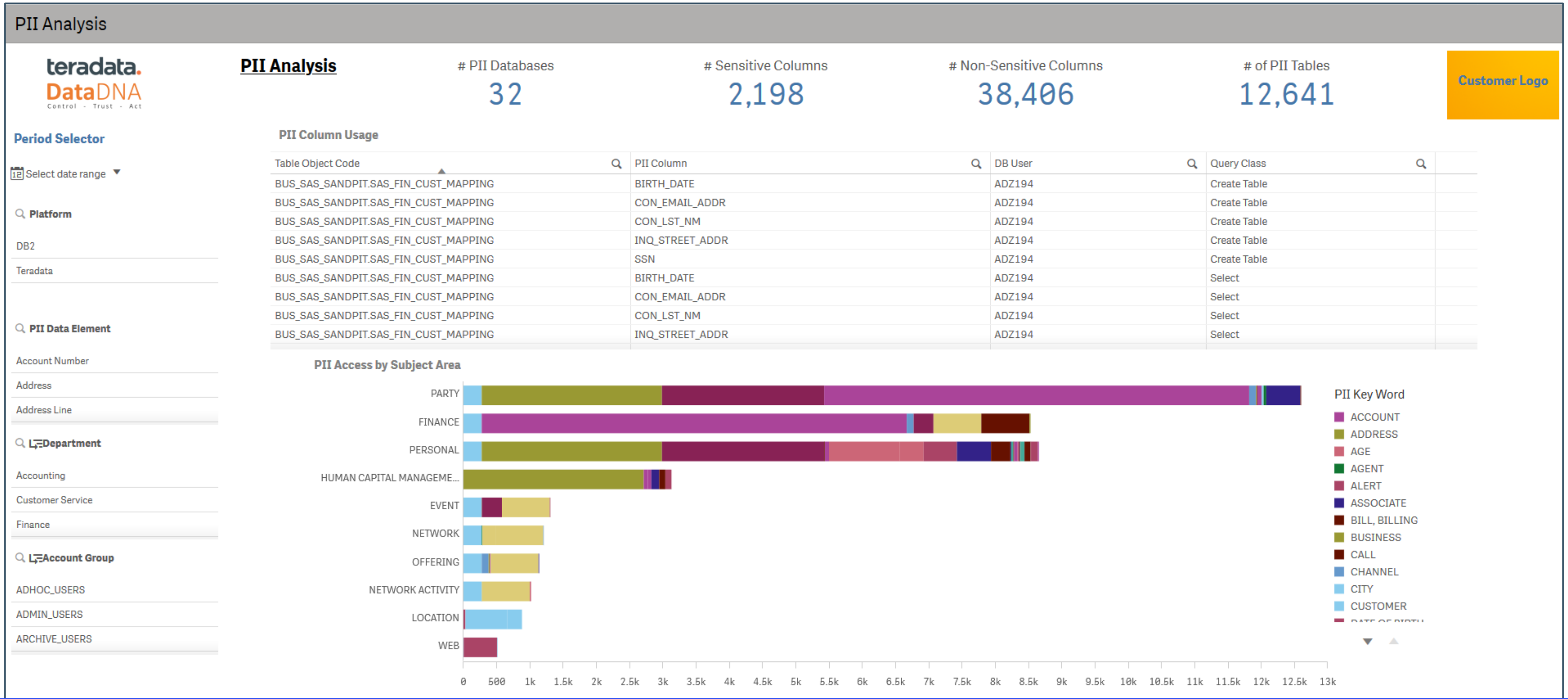

- ✓ Comprehensive PII analysis at column level and the associated subject areas / domain.
- ✓ Ability to analyze types of queries that involve PII data and 'who' are executing those queries.
- ✓ Combining this with 'User Trend Insights' (here) can provide deeper insights on PII – critical for regulatory requirements.

PII Insights

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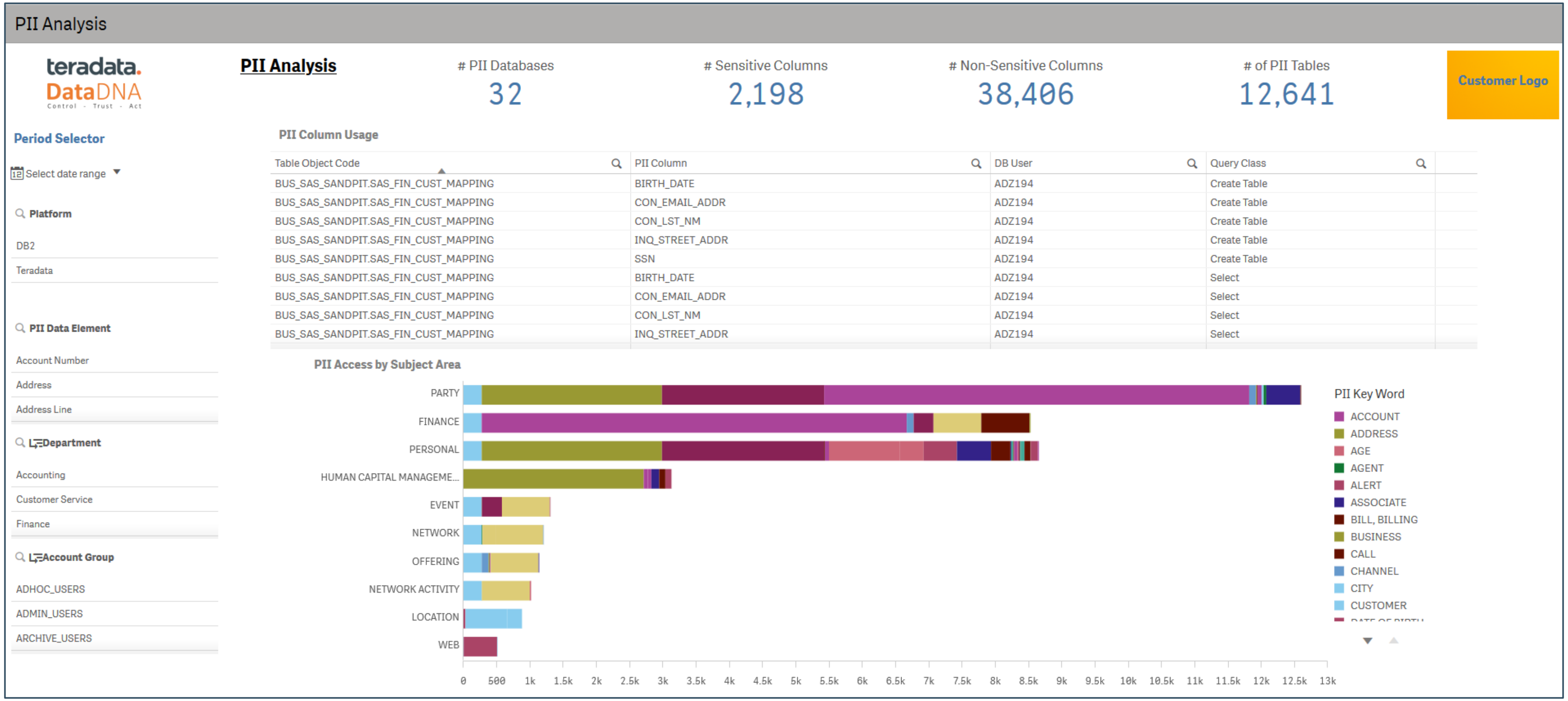

Table Object Code	PII Column	DB User	Query Class
BUS_SAS_SANDPIT.SAS_FIN_CUST_MAPPING	BIRTH_DATE	ADZ194	Create Table
BUS_SAS_SANDPIT.SAS_FIN_CUST_MAPPING	CON_EMAIL_ADDR	ADZ194	Create Table
BUS_SAS_SANDPIT.SAS_FIN_CUST_MAPPING	CON_LST_NM	ADZ194	Create Table
BUS_SAS_SANDPIT.SAS_FIN_CUST_MAPPING	INQ_STREET_ADDR	ADZ194	Create Table
BUS_SAS_SANDPIT.SAS_FIN_CUST_MAPPING	SSN	ADZ194	Create Table
BUS_SAS_SANDPIT.SAS_FIN_CUST_MAPPING	BIRTH_DATE	ADZ194	Select
BUS_SAS_SANDPIT.SAS_FIN_CUST_MAPPING	CON_EMAIL_ADDR	ADZ194	Select
BUS_SAS_SANDPIT.SAS_FIN_CUST_MAPPING	CON_LST_NM	ADZ194	Select
BUS_SAS_SANDPIT.SAS_FIN_CUST_MAPPING	INQ_STREET_ADDR	ADZ194	Select

- ✓ Comprehensive PII analysis at column level and the associated subject areas / domain.
- ✓ Ability to analyze types of queries that involve PII data and ‘who’ are executing those queries.
- ✓ Combining this with ‘User Trend Insights’ (here) can provide deeper insights on PII – critical for regulatory requirements.

Executive Summary / Actions

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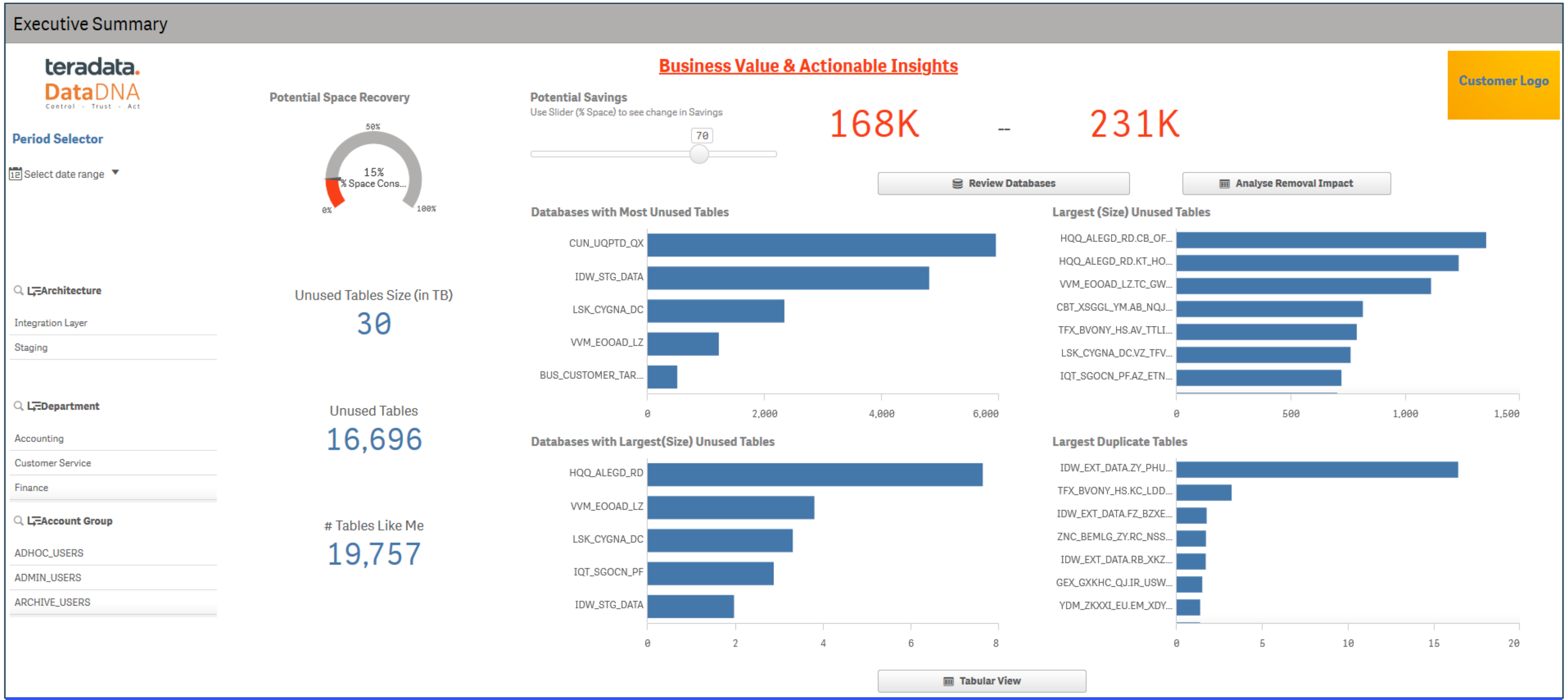

- ✓ Summarized view of the actionable insights e.g. Largest unused tables, Potential Duplicates with ability to understand associated impact analysis.
- ✓ Optional view of potential savings – requires additional metadata/information about the platform to enable this feature.

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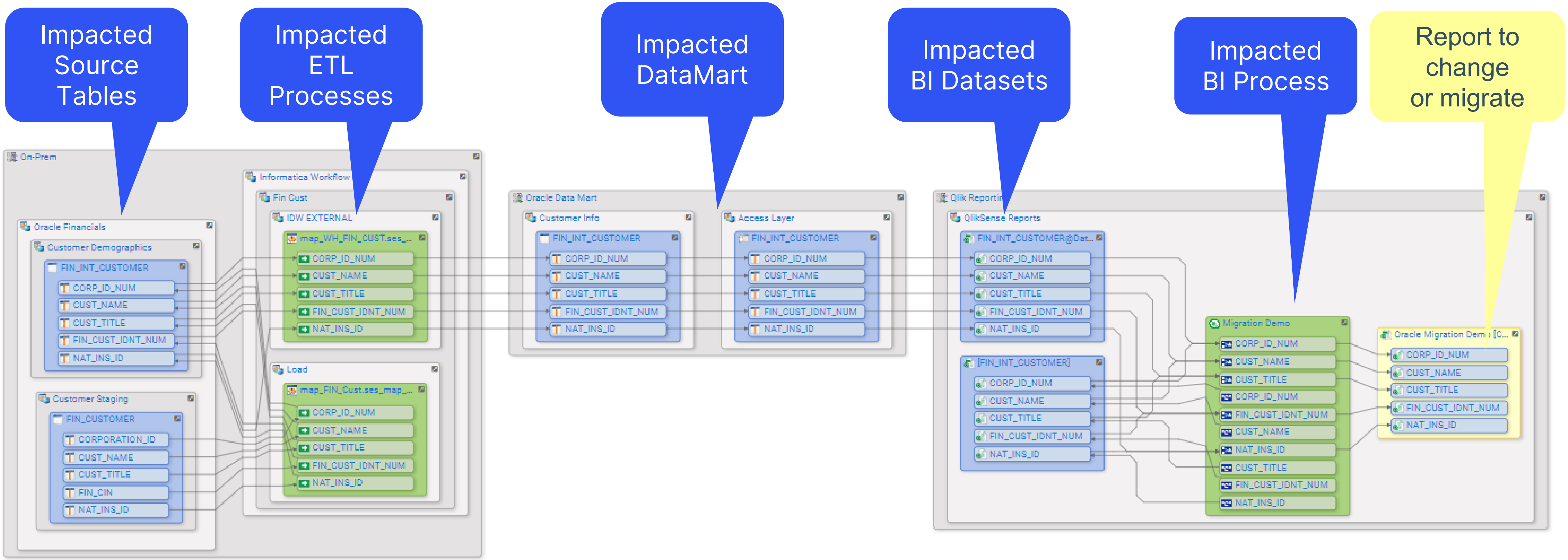
Impact of Change Analysis

DataDNA Usage Analytics

Migration of Legacy Report – Lineage tell us all the impacted objects



- ✓ Fact based lineage provides details of every table/view/file and process that would impacted by a change
- ✓ This example shows every upstream object that is the legacy report is dependent on
- ✓ Impact analysis is built on DataDNA’s column level lineage and usage analytics

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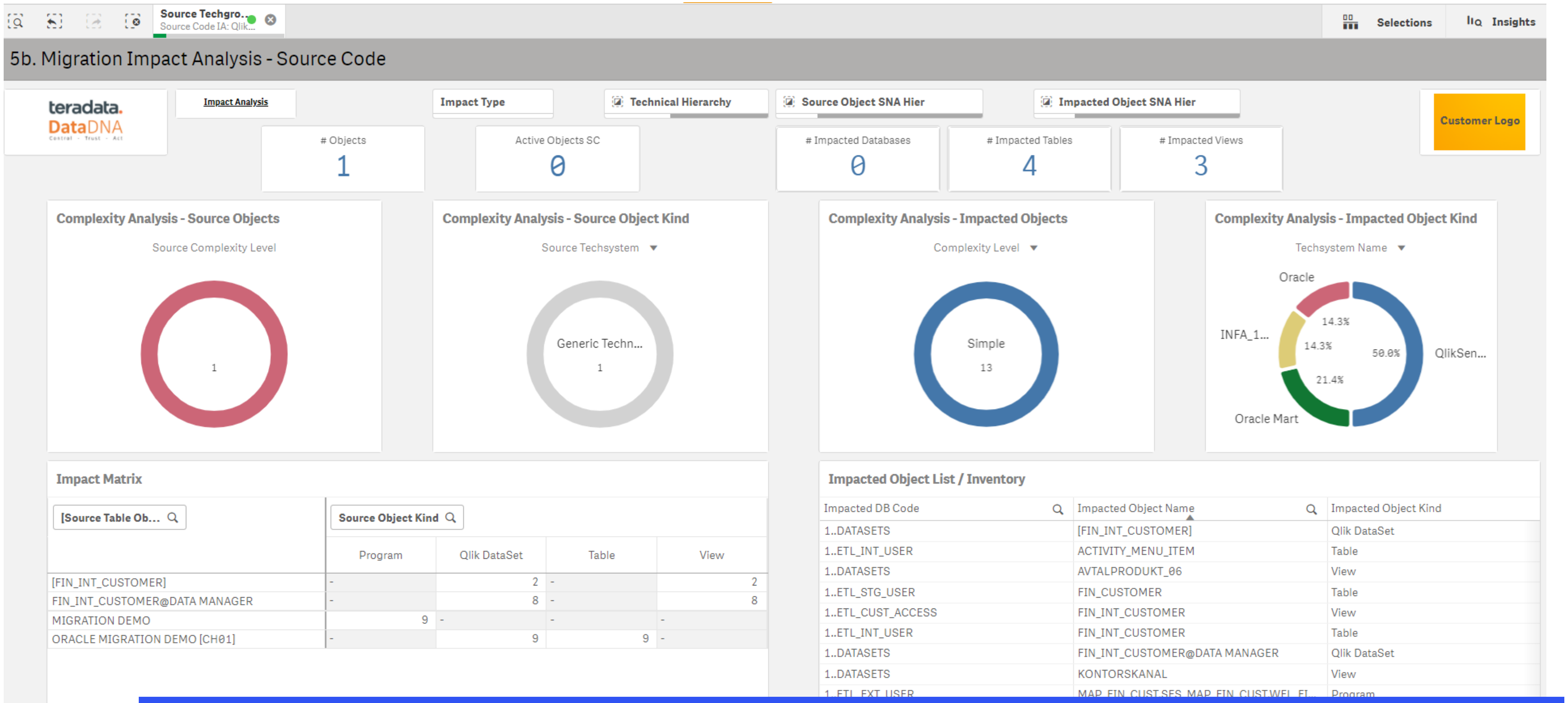
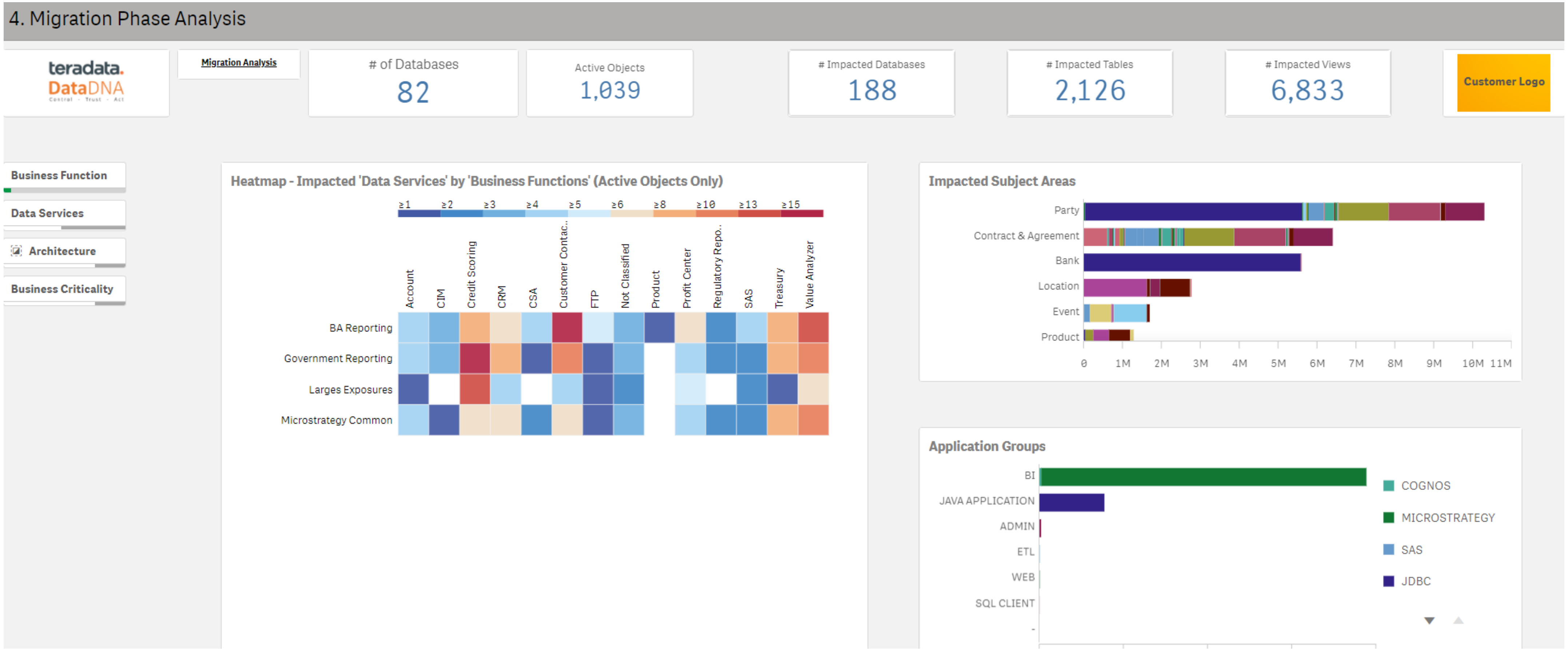
Migration of Legacy Report – Impact Analysis Qlik Report



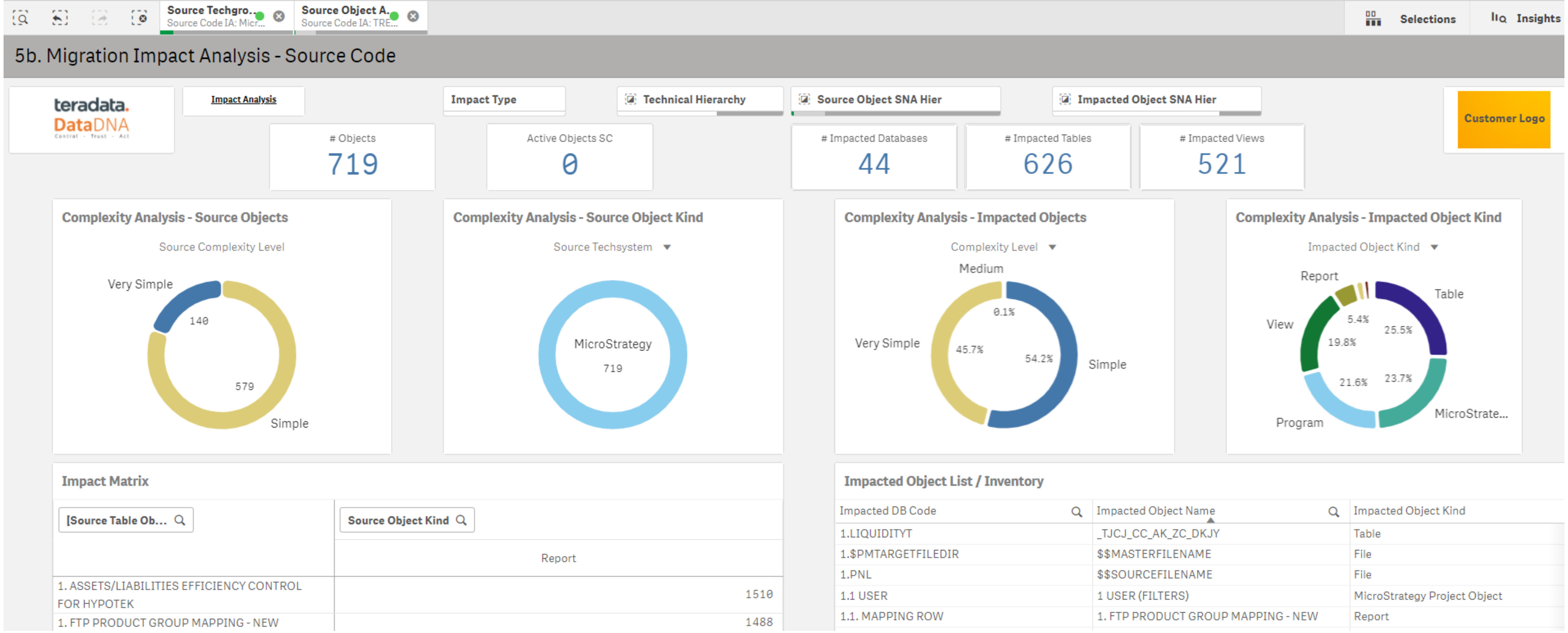
- ✓ Impact analysis is built on DataDNA's column level lineage and usage analytics
- ✓ This dashboard details all the impacted objects for the one legacy report
- ✓ It provides all the information developers need to migrate the legacy report

Impact Analysis for Business Area – Regulatory reporting



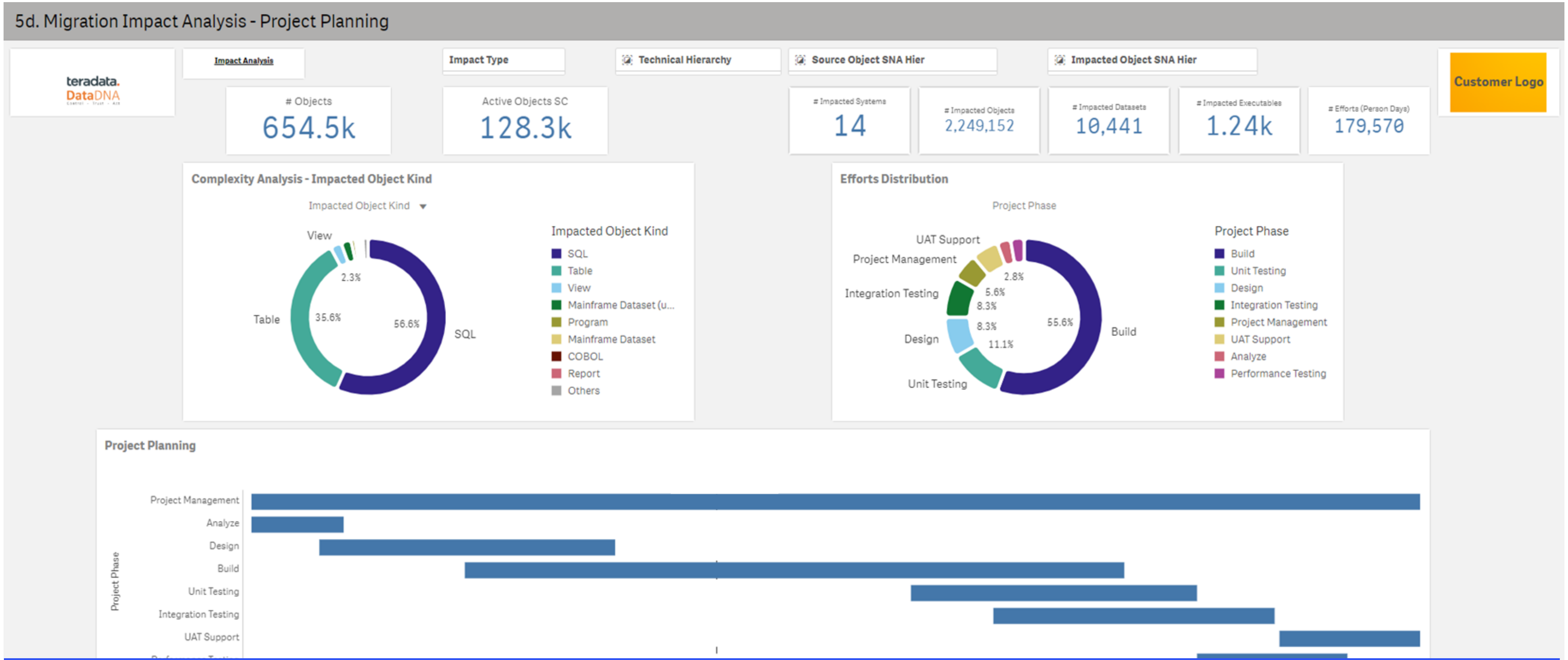
- ✓ Impact analysis can show the impact of change to business areas applications and users
- ✓ This example shows the impact of migrating a business area and function

Impact Analysis for MicroStrategy Reporting



- ✓ This example shows the impact of migrating every report in the MicroStrategy reporting environment
- ✓ The number of impacted datasets and programs across Teradata, informatica and the mainframe
- ✓ Even the complexity of the code is automatically identified by DataDNA

Migration Planning - Project Plan



- ✓ Impact Analysis enabling Project Planning
- ✓ Complexity analytics driving planned effect to complete each stage of the project (Analysis, Design & Build)
- ✓ Multi-planned

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Column Level Impact Analysis (including transformations)

DataDNA Usage Analytics

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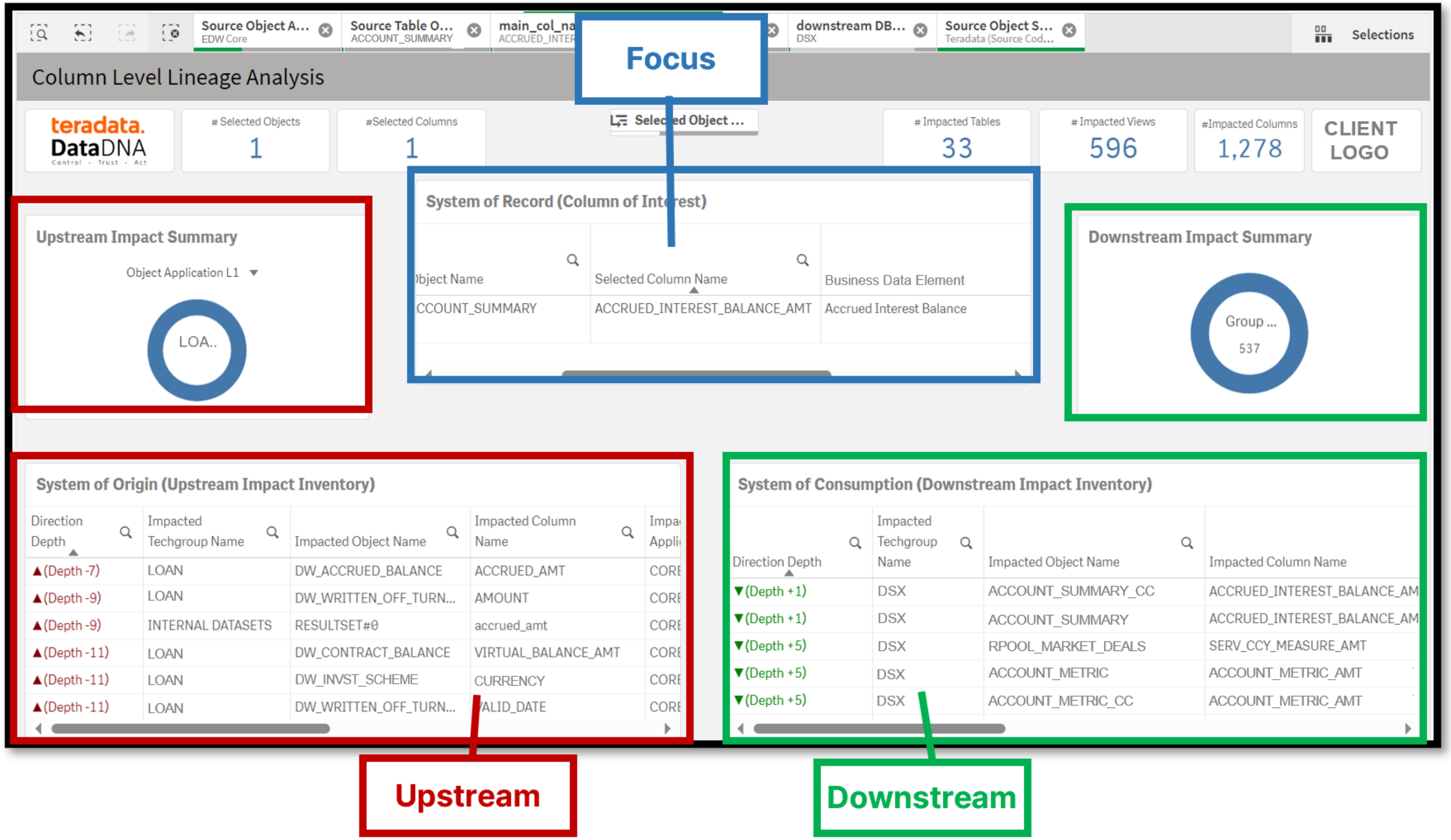
Lineage Analysis – Supports Data Lineage Navigation & Filtering

- ✓ Impact analysis simplifies navigation of data lineage in a complex data ecosystem
- ✓ It can be used for various analysis on top of fact-based data lineage.
- ✓ Hierarchies and filters to help finding relevant data assets, their dependencies and transformation rules.



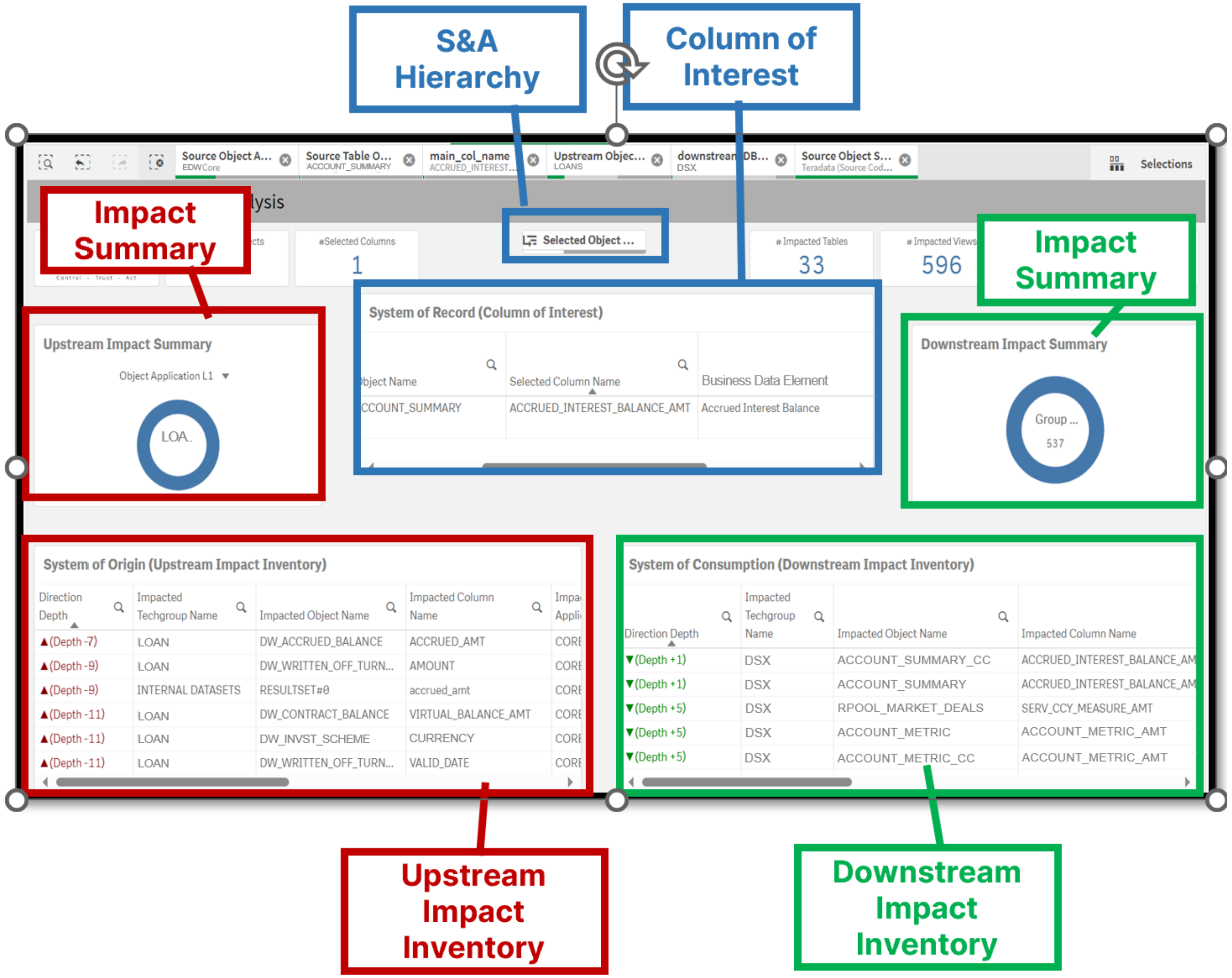
Column Level Lineage Analysis – Overview of Filters

General Filters – based on ingested metadata and hierarchies

- ✓ System & Application (S&A) Hierarchy
- ✓ Column of Interest
- ✓ Impact summaries
- ✓ Impact Inventory

Business Data Element (BDE) Filters - based on ingested BDE to EDW column mappings

- ✓ Used with Column of Interest



- ✓ Filters help to identify the Column of Interest and the change impacts Upstream & Downstream

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Sheet Column Level Lineage Analysis

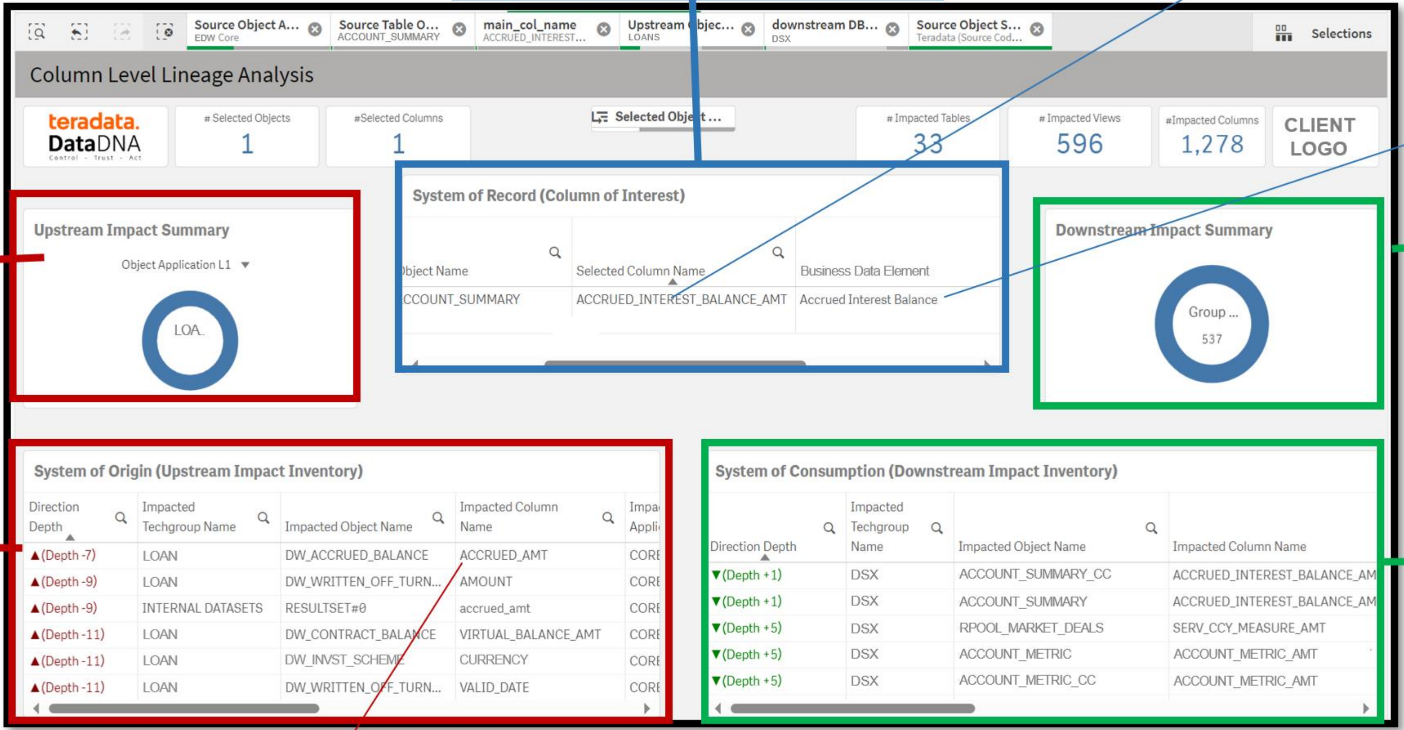
Column of Interest that is in the focus of the analysis

ACCOUNT_SUMMARY.ACCRUED_INTEREST_BALANCE_AMT

Filtering with BDEs is possible

Loans used as a filter for **upstream** flow

DSX used as a filter for **downstream** flow

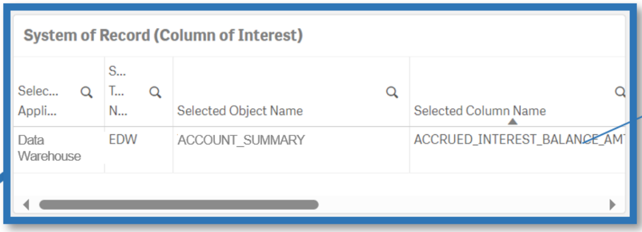


Upstream impacts from the Column of Interest

Downstream impacts from the Column of Interest

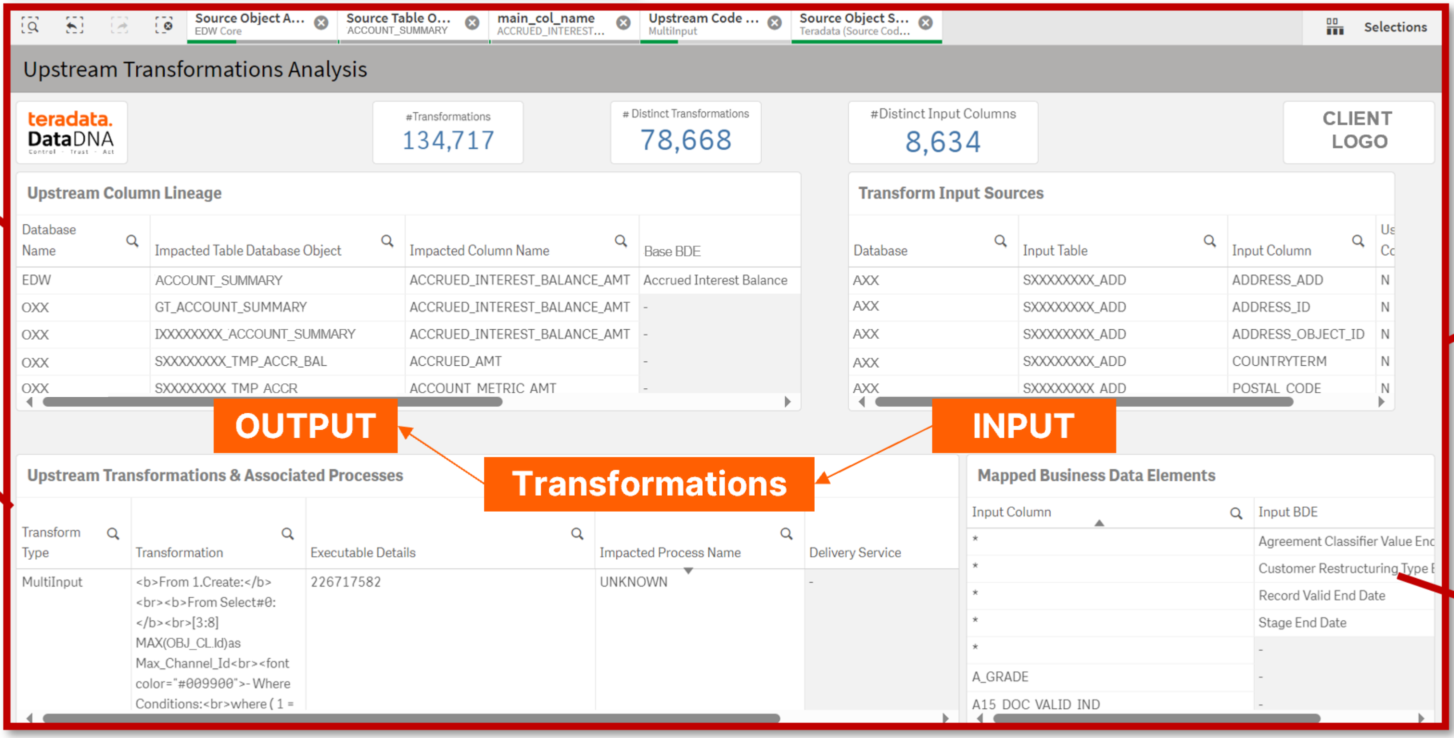
Loans DW_ACCRUED_BALANCE

Sheet Upstream Transformations Analysis



ACCOUNT_SUMMARY.ACCRUED_INTEREST_BALANCE_AMT

Object & Column of interest have been selected in previous sheets



Upstream Impacted objects & columns

All Upstream transformations for the selected column

Helps to identify the hardest code blocks that are impacted

or where the main business rules are applied

Identifies data sources for the transformations

If selected column relates to BDE's they are listed here

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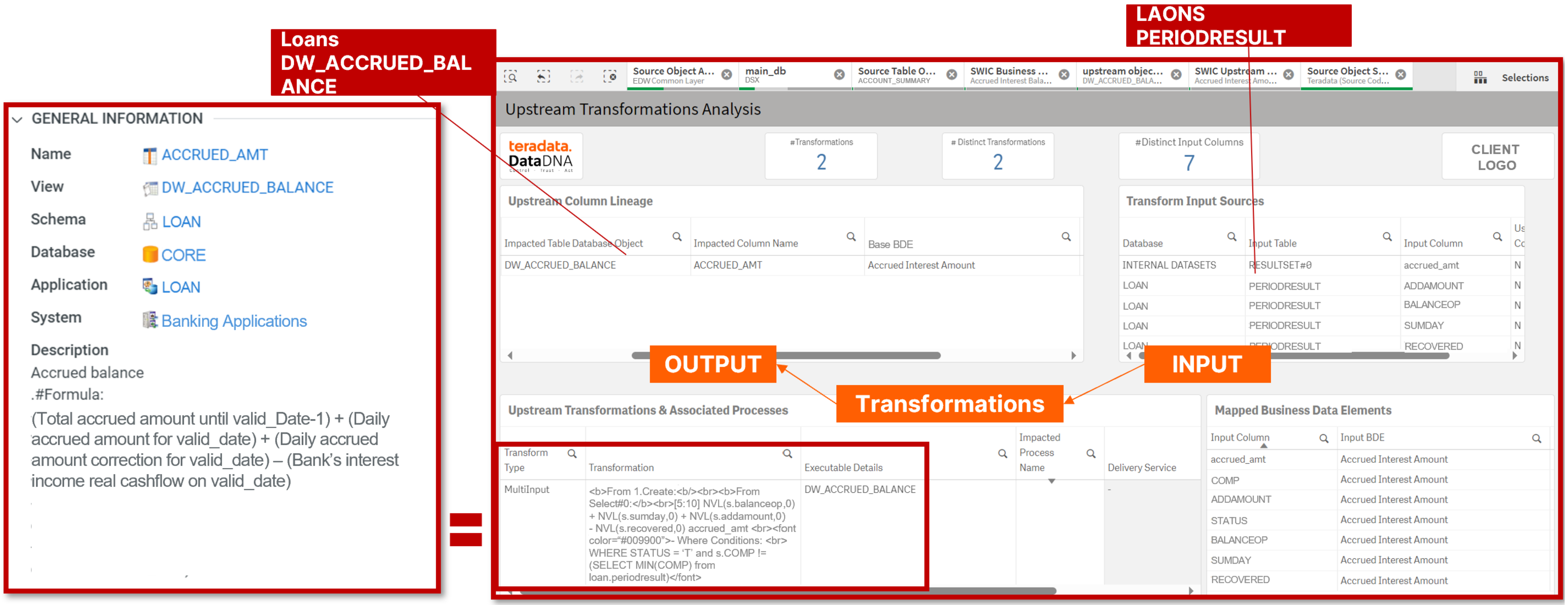
Loans - DW_ACCRUED_BALANCE.Accrued_Amt



Business Value Frameworks

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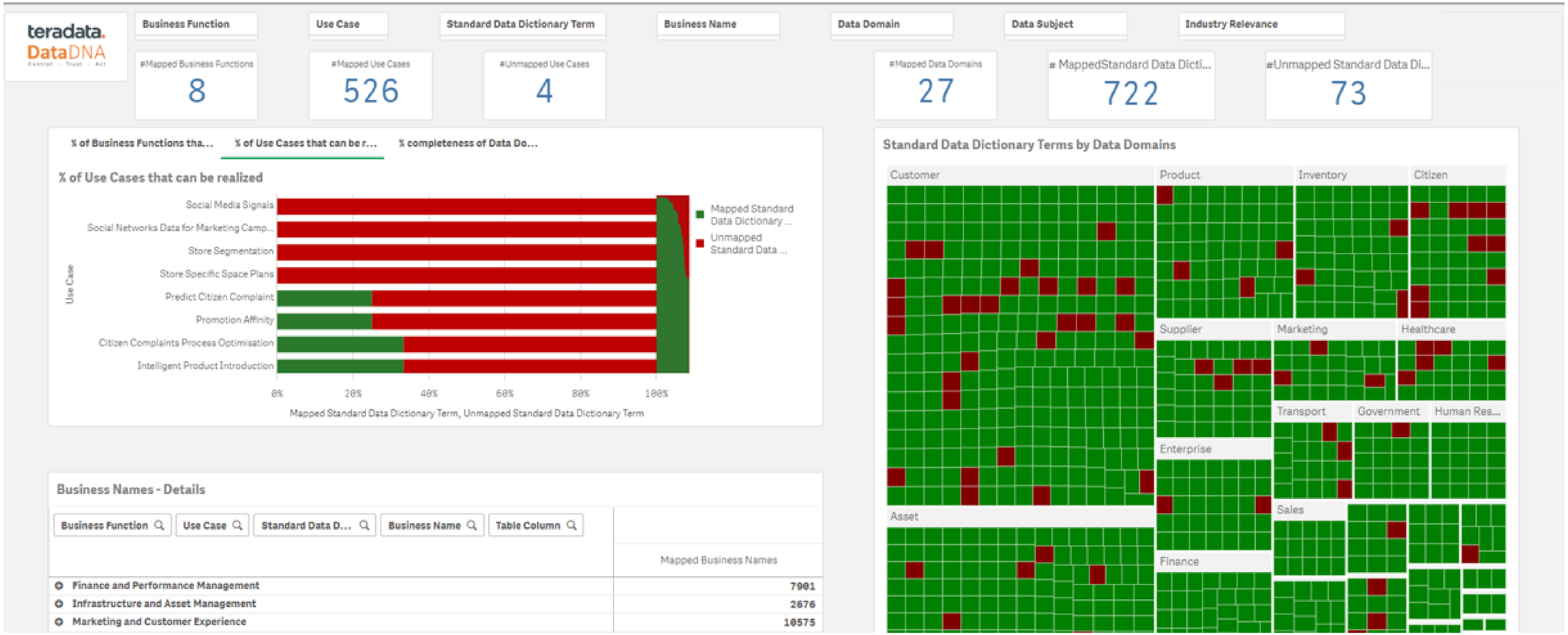
EDW Roadmap



- ✓ Automated machine learning mapping of business terms to physical columns.
- ✓ Data Domain assessment, data missing from the warehouse shown in red
- ✓ What use case can be supported from the data available in the warehouse and what data is needed to support the others.

Query consumption analytics

DataDNA Usage Analytics

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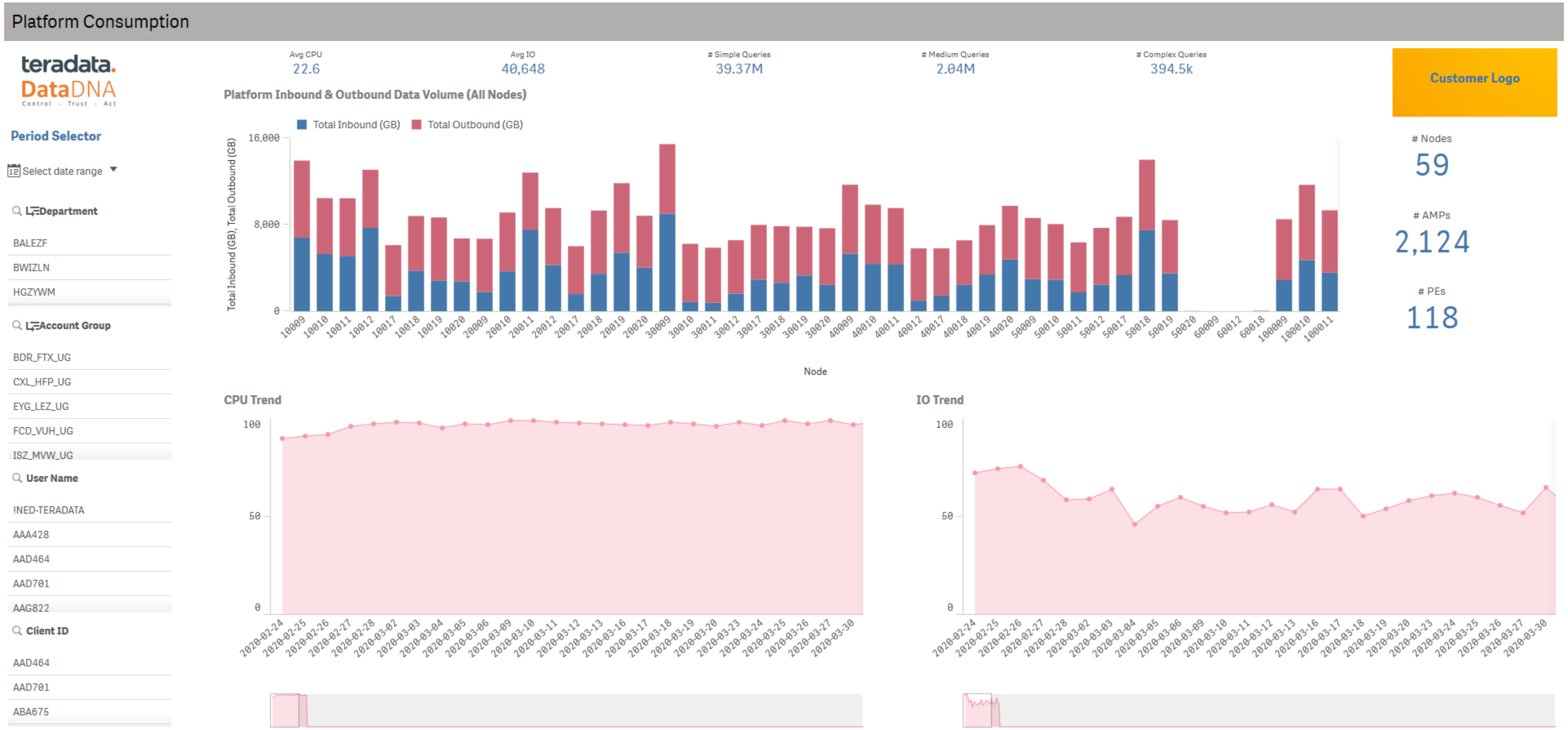
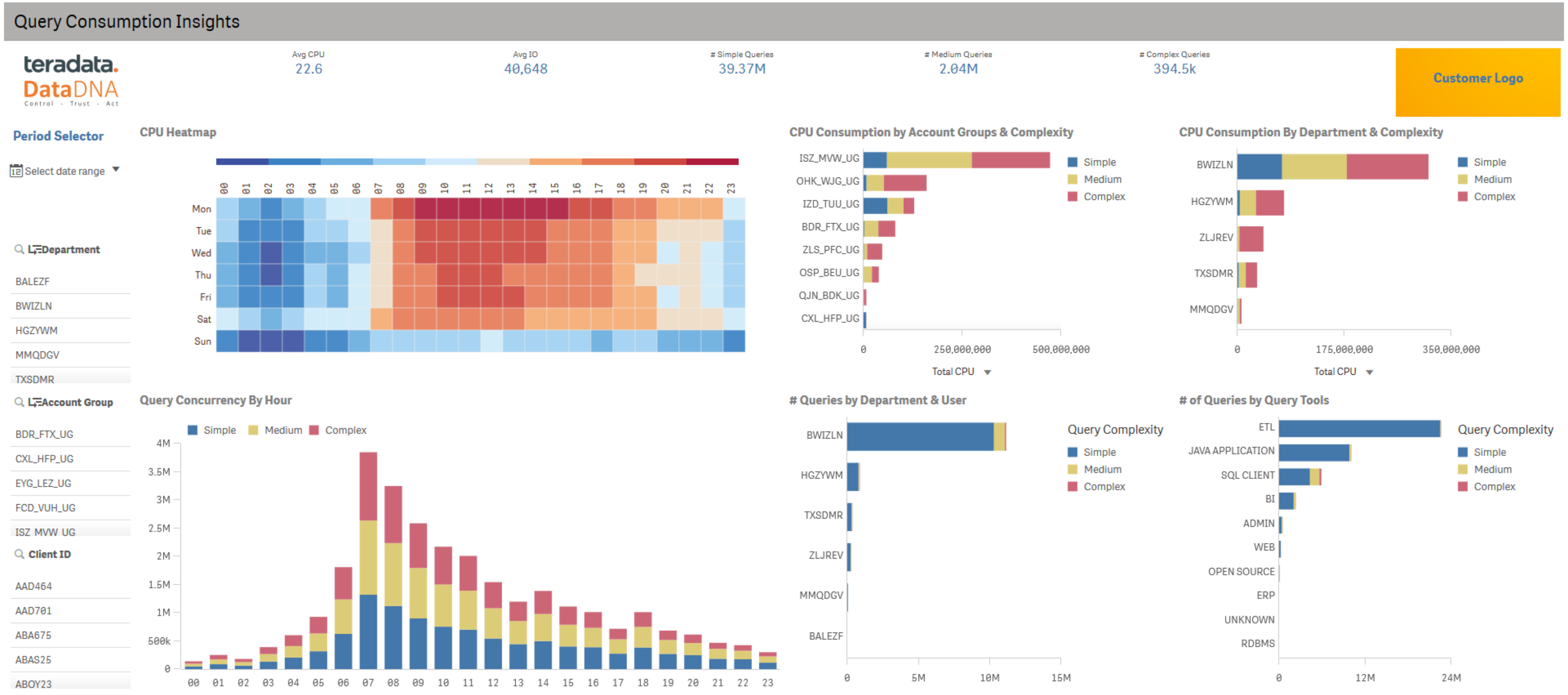
Platform Consumption



- ✓ Summarized view of inbound and outbound data volumes on the platform (Teradata Only)
- ✓ CPU and IO Trend Analysis over the analysis time period.

Query Consumption Insights



- ✓ Insights on CPU and IO consumptions by Query Complexity, Departments & Account Groups – requires departmental information to be supplied to enable this feature.
- ✓ CPU Heatmap to indicate busy periods and hourly query concurrency by Query Complexity.

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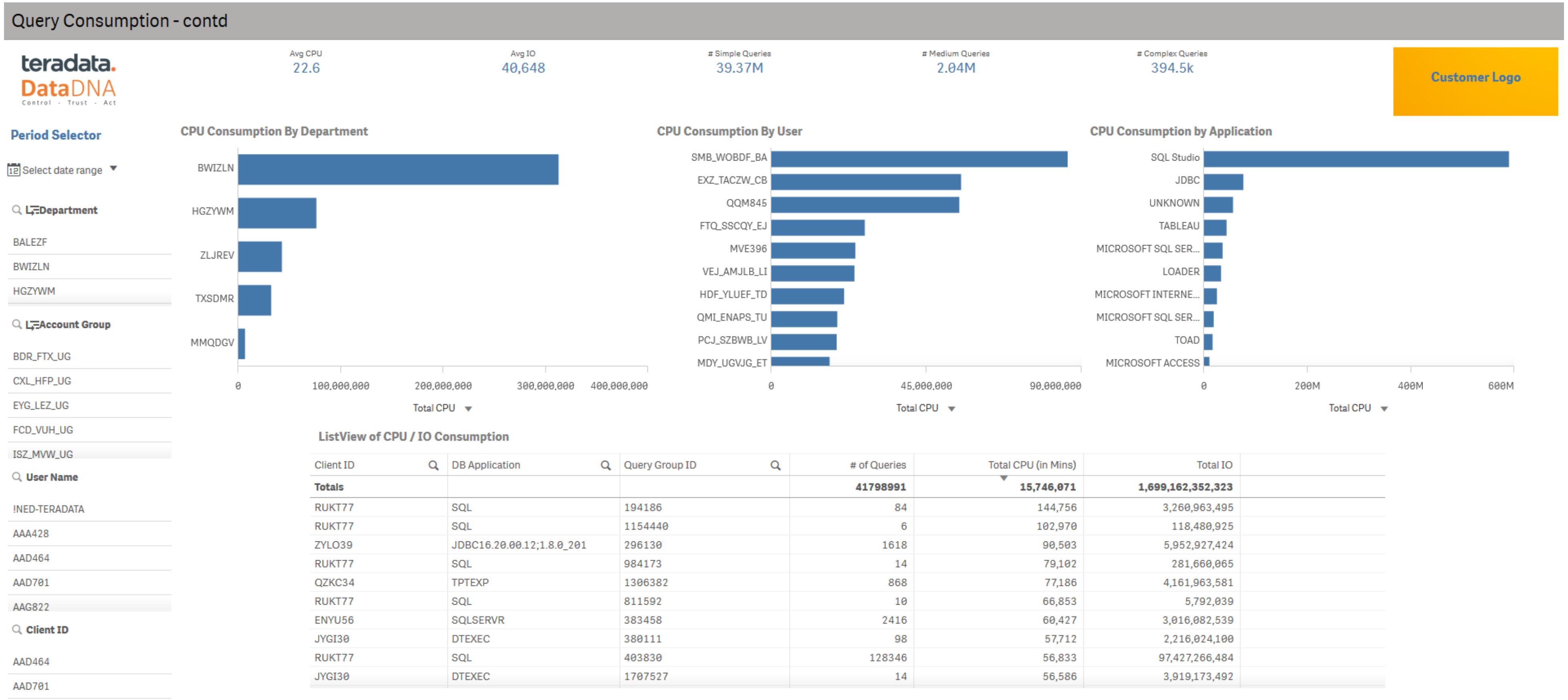
Query Consumption Insights



Client ID	DB Application	Query Group ID	# of Queries	Total CPU (in Mins)	Total IO
Totals			41798991	15,746,071	1,699,162,352,323
RUKT77	SQL	194186	84	144,756	3,260,963,495
RUKT77	SQL	1154440	6	102,970	118,480,925
ZYLO39	JDBC16.20.00.12;1.8.0_201	296130	1618	90,503	5,952,927,424
RUKT77	SQL	984173	14	79,102	281,660,065
QZKC34	TPTEXP	1306382	868	77,186	4,161,963,581
RUKT77	SQL	811592	10	66,853	5,792,039
ENYU56	SQLSERVR	383458	2416	60,427	3,016,082,539
JYGI30	DTEXEC	380111	98	57,712	2,216,024,100
RUKT77	SQL	403830	128346	56,833	97,427,266,484
JYGI30	DTEXEC	1707527	14	56,586	3,919,173,492

- ✓ Provides ability to drill down into CPU and IO consumption by Department & User.
- ✓ Detailed view of # of queries by applications
- ✓ Client ID analysis and associated client applications using ‘which’ account ids. i.e. identify access outliers

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DataDNA - Single solution
Covering ALL aspects of efficient management of data assets

DataDNA
Control - Trust - Act

COLLABORATION
- Data Catalog / Business Glossary ✓
- Metadata Driven Development ✓
- Data Quality Rules ✓
- Reference Data ✓
- Collaboration & Workflows ✓

AUTOMATION
- Usage Analytics ✓
- Inference Engine ✓
- Impact Analysis ✓
- Vertical Lineage ✓
- Horizontal Lineage ✓
- Data Dictionary ✓

Collaboration & fact-based decisions on automatically populated repository

Top Down

Bottom Up

Automated Metadata Discovery

Automation enables business views linked to physical evidence
–
Foundation for collaboration & Fact-Based Data Governance

DataDNA Collaborative Features and Requirements

Collaboration & fact-based decisions on automatically populated repository

Top Down

Data Catalog / Busines Glossary	Strong functionality for managing data catalogue, glossaries, term relationships and hierarchies across functions
Metadata Driven Development	Creating new assets & solutions by optimally reusing the discovered metadata
Business Data Quality	Maintenance of rules to secure compliance of regulations and data quality
Reference Data	Extendible & customizable reference data libraries, enable visibility of the business level visibility information assets & analytics
Collaboration & Workflows	Informal & formal accountabilities with workflows & collaboration. Evolving business glossary, rules & handling issues

DataDNA service - unique IP & capability ?

Bottom Up

DataDNA

Prebuilt Service

IP Driven

Metadata Driven

Capability	Provides fact-based insights to support new business use cases & improvements
Usage Analytics	**Provides focus on data assets that have the most value & risk for business usage** • An understanding of who uses what data, when and with what tools. • Platform and Query Analytics
Inference Engine	**Inference Engine** • Automated mapping of PII and business glossary terms based on column level lineage
Impact Analysis	**Impact Analysis** • Full upstream and downstream impact of any table, database, application of data warehouse
Vertical Lineage	**Semantic Discovery using business vocabularies** • Leveraging customer business metadata, conceptual & logical data models - **top-down** **Automated, Accurate, Auditable & Repeatable Vertical Lineage** • Relating business and technical metadata is automatically inferenced from metadata relationships in (Horizontal) data lineage - **bottom-up**
Horizontal Lineage	**Automated Usage Lineage – based on Usage** • Shows the data processing steps that were ACTUALLY EXECUTED between source and target including end user computing – Reveals shadow IT **Automated Physical Lineage – based on Source Code** • Shows the formally implemented data flows cross platform & column level
Data Dictionary	**Automated Technical metadata ingestion** from variety of data sources • Ability to handle history of changes. Scales to large metadata sets.

Insights

- Cross platform – preferred source analysis
- Understand where to apply Data Quality
- GDPR – Personal (PII) data usage
- Duplicate table analysis
- Data mart/ sandbox overlap analysis
- Migration/ change impact planning
- Enable data remediation efforts with least effort and most impact

DataDNA service – Capability List

DataDNA

Control - Trust - Act

			Metadata Sources											
				Database Catlog						Organisation Structures			Model	
			Business Glossary	Tables	Views	Columns	Usage / Queries	Source Code	Appli-cation	Users	Business	Architecture & Design	Physical	Logical
DataDNA Deliverables	Usage Analytics Portal	Core Object Definitions		✓	✓	✓			✓			✓	✓	✓
		Organisational Hierarchies												
		Reporting Hierarchies							✓			✓		
		Disk Space Utilisation		✓					✓			✓		
		Database Metrics		✓					✓			✓		
		Database Read/Write Analysis		✓	✓	✓	✓		✓			✓		
		System and Application Hierarchies							✓			✓		
		Unused Tables		✓	✓		✓		✓			✓		
		Table Usage		✓			✓		✓			✓		
		Table Metrics		✓					✓			✓		
		Table Size		✓					✓			✓		
		Query Parsing		✓		✓	✓					✓		
		Query Classification					✓					✓		
		Platform Performance & Optimisation					✓		✓			✓		
		Duplication - Tables Like Me		✓		✓			✓			✓		
		Table Cluster Analysis		✓					✓			✓		
		Subject Area Analysis		✓		✓	✓		✓			✓		
		User Trend Analysis										✓		
		Personal Data usage (PII)		✓	✓	✓								
		Broken Views		✓	✓							✓		
		Platform Consumption					✓					✓		
		Query Consumption Insights		✓	✓		✓					✓		
		User Query Activity		✓	✓		✓					✓		
		Feature Insights - Teradata										✓		
		Impact Assessment		✓	✓				✓			✓		
	Lineage Portal	Table/Column Level Lineage		✓	✓	✓			✓			✓	✓	✓
		Source Code Lineage							✓			✓		
		Process Lineage												
		Personal Data Annotations (PII)		✓	✓	✓			✓					
		Personal Data Inferred via lineage (PII)		✓	✓	✓			✓					
		Usage Lineage		✓	✓	✓	✓							
		Impact Assessment (through lineage)		✓	✓	✓			✓					
		Physical / Logical Model Mapping											✓	✓
		Logical Model Annotations											✓	✓
		Business Glossary Annotations	✓										✓	✓
		Business Glossary Management	✓											

Thank You

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